Appendix C – Flow Calculations

Design Settings

Rainfall Methodology	FSR	Maximum Time of Concentration (mins)	30.00
Return Period (years)	2	Maximum Rainfall (mm/hr)	50.0
Additional Flow (%)	0	Minimum Velocity (m/s)	1.00
FSR Region	England and Wales	Connection Type	Level Soffits
M5-60 (mm)	20.000	Minimum Backdrop Height (m)	0.200
Ratio-R	0.400	Preferred Cover Depth (m)	1.200
CV	0.750	Include Intermediate Ground	✓
Time of Entry (mins)	5.00	Enforce best practice design rules	✓

Nodes

Name	Area (ha)	T of E (mins)	Cover Level (m)	Diameter (mm)	Easting (m)	Northing (m)	Depth (m)
6	0.000	10.00	92.424	1350	419574.518	187451.974	1.340
7	0.078	10.00	92.809	1350	419618.192	187490.602	1.986
8	0.660	10.00	92.472	1350	419654.412	187522.847	1.765
30	0.230	15.00	92.415	1350	419670.494	187538.459	1.646
9	0.023	10.00	92.376	1350	419670.661	187529.285	1.786
10	0.041	10.00	92.589	1500	419694.470	187519.011	2.056
40	0.198	15.00	92.418	1200	419744.787	187435.344	1.975
11	0.154	10.00	92.316	1500	419751.281	187460.113	2.037
50	0.265	15.00	92.277	1200	419793.986	187454.517	1.737
S79	0.000	5.00	92.406	1500	419772.361	187450.351	2.167
12	0.016	10.00	92.233	1500	419794.998	187444.692	2.034
S78	0.000	5.00	92.089	1500	419808.120	187434.152	1.919
13	0.130	10.00	92.229	1500	419852.097	187376.846	2.182
14	0.057	10.00	92.089	1500	419873.118	187354.472	2.095
15	0.150	10.00	91.943	1800	419890.913	187350.318	2.137
60	0.265	5.00	92.151	1200	419924.454	187388.859	1.821
16	0.060	5.00	91.945	1800	419935.575	187356.978	2.352
17	0.000		90.525	1800	419954.275	187339.070	1.005
18	0.000	5.00	90.800	1800	419999.287	187314.408	
Basin A Flow Control	0.000	5.00	90.982	1800	420139.953	187253.093	1.791
Basin A Outfall			89.837	1350	420222.346	187274.600	0.947
Basin A to Swale			90.250		420173.163	187242.871	1.144
S87			90.250		420196.762	187230.967	1.170
S88			90.250		420218.697	187228.818	1.192
S89			90.271		420247.892	187253.643	1.308
S92			90.270		420244.486	187263.216	1.317
S90			90.270		420244.260	187272.558	1.326
S12	0.052	15.00	98.992	1350	419219.225	187158.445	1.425
S13	0.052	15.00	99.134	1200	419212.887	187191.085	1.714
S14	0.052	15.00	98.542	1350	419212.734	187239.424	1.500
S15	0.180	15.00	98.094	1350	419199.040	187270.760	1.575
S16	0.180	15.00	97.525	1350	419176.176	187307.905	1.655
S41	0.128	15.00	99.256	1350	419051.014	187353.832	1.753
S42	0.128	15.00	98.810	1350	419086.623	187378.839	1.650
S43	0.128	15.00	98.671	1350	419096.534	187373.769	1.661
S37	0.128	15.00	99.654	1350	419072.025	187262.568	1.500
S38	0.128	15.00	99.081	1350	419107.438	187284.912	1.500
S39	0.128	15.00	98.868	1350	419110.745	187298.140	1.518
S40	0.128	15.00	98.372	1200	419104.414	187334.303	1.652
S17	0.180	15.00	97.024	1350	419163.023	187344.880	1.776

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Nodes

Name	Area (ha)	T of E (mins)	Cover Level (m)	Diameter (mm)	Easting (m)	Northing (m)	Depth (m)
S18	0.180	15.00	96.680	1350	419162.434	187371.928	1.883
S35	0.128	15.00	96.947	1350	419136.639	187438.587	1.500
S36	0.128	15.00	96.417	1350	419163.393	187402.742	1.575
S19	0.052	15.00	96.334	1350	419172.119	187396.148	1.787
S34	0.071	15.00	95.967	1350	419180.260	187464.095	1.425
S20	0.123	15.00	95.729	1350	419208.265	187428.759	1.650
S21	0.123	15.00	95.502	1350	419221.439	187439.571	2.039
S22	0.123	15.00	95.351	1350	419221.523	187450.544	2.036
S23	0.123	15.00	94.986	1350	419207.703	187475.315	2.041
S24	0.123	15.00	94.821	1500	419212.033	187487.655	2.129
S25	0.123	15.00	93.931	1500	419269.702	187526.762	2.211
S26	0.109	15.00	92.914	1800	419336.421	187570.606	2.094
S27	0.109	15.00	92.753	1800	419352.227	187566.780	2.090
S28	0.109	15.00	93.042	1800	419392.176	187535.233	2.509
S29	0.212	15.00	93.392	1800	419416.168	187494.330	2.925
S30	0.212	15.00	93.044	1800	419451.502	187457.486	2.647
S31	0.212	15.00	92.581	1800	419477.685	187404.320	2.341
S32	0.212	15.00	92.337	1800	419512.697	187374.351	2.169
S44	1.000	15.00	96.099	1350	419413.187	187266.592	1.498
S45	0.217	10.00	94.537	1200	419472.386	187308.367	1.807
S46	0.217	10.00	93.256	1200	419513.739	187350.990	2.026
S33	0.057	5.00	92.852	1800	419524.515	187365.649	2.702
S47	0.217	5.00	92.151	1350	419544.212	187393.468	2.042
S48	0.057	15.00	92.235	1200	419560.216	187432.111	2.178
S49			93.000	1350	419591.449	187414.323	3.009
S58	0.190	10.00	93.000	1350	419569.462	187391.824	2.480
S61	0.190	10.00	93.000	1350	419572.793	187285.223	2.524
S59 Flow Control	0.190	10.00	93.000	1350	419604.332	187324.243	2.625
S60			92.500	1350	419645.030	187334.273	2.605
S50			92.500	1350	419719.115	187223.779	2.738
S55			92.500	1350	419750.577	187211.371	2.772
S51			92.500	1350	419796.344	187225.646	2.820
S52			92.500	1350	419868.460	187305.108	2.927
S56			92.500	1350	419887.557	187309.239	2.947
S57			92.500	1350	419935.749	187304.315	2.995
S53			92.500	1350	419993.281	187306.638	3.060
S54 HW			90.600	1900	420013.487	187309.882	1.372
S62 HW		5.00	90.495	1350	420130.841	187257.374	1.270
S93			90.600		419995.735	187318.516	1.126
S73	0.128	5.00	91.846	1350	419680.300	187456.316	1.373
S74	0.128	5.00	92.097	1350	419706.258	187410.625	1.845
S71	0.128	5.00	91.692	1350	419665.486	187384.467	1.595
S72	0.104	5.00	92.000		419634.976	187348.394	2.057
S75	0.104	5.00	92.194	1350	419745.058	187416.105	1.741
S76	0.104	5.00	91.893	1350	419766.732	187379.260	1.582
S77	0.104	5.00	91.941	1350	419780.835	187361.395	1.554
S68	0.104	5.00	91.872	1350	419732.848	187358.983	1.693
S69	0.104	5.00	91.902	1350	419695.286	187336.230	1.869
S70	0.104	15.00	92.000		419667.902	187300.161	2.118
S65	0.104	15.00	92.284	1350	419811.543	187371.010	1.983
S66	0.104	15.00	92.035	1350	419792.856	187350.144	1.927

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Nodes

Name	Area (ha)	T of E (mins)	Cover Level (m)	Diameter (mm)	Easting (m)	Northing (m)	Depth (m)
S67	0.104	15.00	91.490	1350	419750.450	187303.108	1.659
S62	0.080	15.00	91.939	1350	419862.037	187340.843	2.119
S63	0.080	15.00	91.754	1350	419848.374	187324.690	1.974
S64	0.080	15.00	91.363	1350	419815.658	187288.265	1.493

Links

Name	US Node	DS Node	Length (m)	ks (mm) / n	US IL (m)	DS IL (m)	Fall (m)	Slope (1:X)	Dia (mm)	T of C (mins)	Rain (mm/hr)
1.005	6	7	58.306	0.600	91.084	90.898	0.186	313.5	375	11.59	46.4
1.006	7	8	48.494	0.600	90.823	90.707	0.116	418.1	450	12.64	44.2
1.007	8	9	17.478	0.600	90.707	90.665	0.042	416.1	450	12.78	43.9
2.000	30	9	9.176	0.600	90.769	90.740	0.029	316.4	375	15.19	39.8
1.008	9	10	25.931	0.600	90.590	90.533	0.057	454.9	525	15.67	39.0
1.009	10	11	81.447	0.600	90.533	90.354	0.179	455.0	525	17.01	37.2
1.010	11	S79	23.231	0.600	90.279	90.239	0.040	580.8	600	17.87	36.1
1.211	S79	12	23.334	0.600	90.239	90.199	0.040	583.4	600	18.26	35.6
1.011	12	S78	16.831	0.600	90.199	90.170	0.029	580.4	600	19.31	34.4
5.001	50	12	9.877	0.600	90.540	90.499	0.041	240.9	300	15.13	39.9
1.212	S78	13	72.235	0.600	90.170	90.047	0.123	587.3	600	20.52	33.1
1.012	13	14	30.700	0.600	90.047	89.994	0.053	579.2	600	21.03	32.6
1.013	14	15	18.273	0.600	89.994	89.956	0.038	480.9	600	21.30	32.4
1.014	15	16	45.156	0.600	89.806	89.743	0.063	716.8	750	22.03	31.7
5.000	60	16	33.765	0.600	90.330	90.183	0.147	229.7	300	5.56	50.0
1.015	16	17	16.393	0.600	89.593	89.520	0.073	224.6	900	15.41	30.5
1.016	17	S93	74.839	0.600	89.520	89.474	0.046	1626.9	1050	16.41	29.4
16.014	S93	S54 HW	19.740	0.600	89.474	89.410	0.064	308.4	525	16.67	37.6

Name	Vel (m/s)	Cap (l/s)	Flow (l/s)	US Depth (m)	DS Depth (m)	Σ Area (ha)	Σ Add Inflow (l/s)	Pro Depth (mm)	Pro Velocity (m/s)
1.005	1.018	112.4	0.0	0.965	1.536	0.000	0.0	0	0.000
1.006	0.988	157.1	9.3	1.536	1.315	0.078	0.0	73	0.548
1.007	0.990	157.5	87.8	1.315	1.261	0.738	0.0	240	1.016
2.000	1.013	111.9	24.8	1.271	1.261	0.230	0.0	120	0.820
1.008	1.043	225.9	104.8	1.261	1.531	0.991	0.0	251	1.024
1.009	1.043	225.8	104.1	1.531	1.437	1.032	0.0	250	1.022
1.010	1.003	283.6	116.0	1.437	1.567	1.186	0.0	267	0.955
1.211	1.001	283.0	114.4	1.567	1.434	1.186	0.0	265	0.950
1.011	1.003	283.7	136.8	1.434	1.319	1.467	0.0	294	0.995
5.001	1.008	71.3	28.7	1.437	1.434	0.265	0.0	132	0.955
1.212	0.997	282.0	131.7	1.319	1.582	1.467	0.0	288	0.981
1.012	1.004	284.0	141.2	1.582	1.495	1.597	0.0	299	1.003
1.013	1.104	312.0	145.1	1.495	1.387	1.654	0.0	287	1.084
1.014	1.037	458.3	154.9	1.387	1.452	1.804	0.0	299	0.940
5.000	1.033	73.0	35.9	1.521	1.462	0.265	0.0	149	1.029
1.015	2.086	1327.3	176.0	1.452	0.105	2.129	0.0	218	1.471
1.016	1.313	5033.3	169.7	-0.045	0.076	2.129	0.0	240	0.582
16.014	1.270	274.9	217.1	0.601	0.665	2.129	0.0	354	1.401

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Links

Name	US Node	DS Node	Length (m)	ks (mm) / n	US IL (m)	DS IL (m)	Fall (m)	Slope (1:X)	Dia (mm)	T of C (mins)	Rain (mm/hr)
20.000	S62 HW	Basin A Flow Control	10.068	0.600	89.225	89.191	0.034	296.1	1050	5.08	50.0
1.018	Basin A Flow Control	Basin A to Swale	33.269	0.600	89.191	89.106	0.085	391.4	450	18.37	27.5
20.002	Basin A to Swale	S87	26.431	0.030	89.106	89.080	0.026	1016.6	1000	19.00	34.7
20.003	S87	S88	22.040	0.030	89.080	89.058	0.022	1001.8	1000	19.52	34.2
20.004	S88	S89	38.323	0.030	89.058	88.963	0.095	403.4	450	21.17	32.5
20.005	S89	S92	10.161	0.030	88.963	88.953	0.010	1016.1	1000	21.43	32.2
20.006	S92	S90	9.345	0.030	88.953	88.944	0.009	1038.3	1000	21.66	32.0
20.007	S90	Basin A Outfall	22.009	0.030	88.944	88.890	0.054	407.6	450	22.62	31.2
1.000	S12	S13	33.250	0.600	97.567	97.420	0.147	226.2	225	5.68	66.5
1.001	S13	S14	48.339	0.600	97.420	97.042	0.378	127.9	300	6.62	62.0
1.002	S14	S15	34.198	0.600	97.042	96.594	0.448	76.3	300	7.11	59.9
1.003	S15	S16	43.618	0.600	96.519	95.870	0.649	67.2	375	7.59	58.0
1.004	S16	S17	39.245	0.600	95.870	95.323	0.547	71.7	375	7.94	56.7
2.000_1	S41	S42	43.514	0.600	97.503	97.160	0.343	126.9	300	5.68	66.5
2.001	S42	S43	11.133	0.600	97.160	97.010	0.150	74.2	300	5.84	65.7
2.002	S43	S40	40.244	0.600	97.010	96.720	0.290	138.8	300	6.63	61.9
3.000	S37	S38	41.873	0.600	98.154	97.581	0.573	73.1	300	5.61	66.8
3.001	S38	S39	13.636	0.600	97.581	97.350	0.231	59.0	300	5.77	66.0
3.002	S39	S40	36.713	0.600	97.350	96.720	0.630	58.3	300	6.23	63.7
2.003	S40	S17	59.556	0.600	96.720	95.395	1.325	44.9	300	7.31	59.1
1.005_1	S17	S18	27.054	0.600	95.248	94.872	0.376	72.0	450	8.15	55.9
1.006_1	S18	S19	26.084	0.600	94.797	94.547	0.250	104.3	450	8.37	55.2
4.000	S35	S36	44.729	0.600	95.447	94.917	0.530	84.4	300	5.68	66.5

Name	Vel (m/s)	Cap (l/s)	Flow (l/s)	US Depth (m)	DS Depth (m)	Σ Area (ha)	Σ Add Inflow (l/s)	Pro Depth (mm)	Pro Velocity (m/s)
20.000	1.997	1729.3	0.0	0.220	0.741	0.000	0.0	0	0.000
1.018	1.021	162.4	0.0	1.341	0.694	0.000	0.0	0	0.000
20.002	0.698	2793.9	0.0	0.144	0.170	0.000	0.0	0	0.000
20.003	0.704	2814.5	0.0	0.170	0.192	0.000	0.0	0	0.000
20.004	0.387	61.5	0.0	0.742	0.858	0.000	0.0	0	0.000
20.005	0.670	2345.0	0.0	0.308	0.317	0.000	0.0	0	0.000
20.006	0.663	2319.8	0.0	0.317	0.326	0.000	0.0	0	0.000
20.007	0.385	61.2	0.0	0.876	0.497	0.000	0.0	0	0.000
1.000	0.865	34.4	9.4	1.200	1.489	0.052	0.0	80	0.740
1.001	1.389	98.2	17.5	1.414	1.200	0.104	0.0	85	1.056
1.002	1.801	127.3	25.3	1.200	1.200	0.156	0.0	91	1.416
1.003	2.213	244.4	52.8	1.200	1.280	0.336	0.0	118	1.777
1.004	2.141	236.5	79.3	1.280	1.326	0.516	0.0	149	1.935
2.000_1	1.394	98.6	23.1	1.453	1.350	0.128	0.0	98	1.143
2.001	1.827	129.1	45.6	1.350	1.361	0.256	0.0	123	1.673
2.002	1.332	94.2	64.4	1.361	1.352	0.384	0.0	183	1.431
3.000	1.841	130.1	23.2	1.200	1.200	0.128	0.0	85	1.401
3.001	2.050	144.9	45.8	1.200	1.218	0.256	0.0	116	1.825
3.002	2.063	145.8	66.3	1.218	1.352	0.384	0.0	142	2.017
2.003	2.351	166.2	143.5	1.352	1.329	0.896	0.0	216	2.632
1.005_1	2.399	381.5	241.2	1.326	1.358	1.592	0.0	261	2.533
1.006_1	1.990	316.5	265.1	1.433	1.337	1.772	0.0	317	2.217
4.000	1.712	121.0	23.1	1.200	1.200	0.128	0.0	88	1.327

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Links

Name	US Node	DS Node	Length (m)	ks (mm) / n	US IL (m)	DS IL (m)	Fall (m)	Slope (1:X)	Dia (mm)	T of C (mins)	Rain (mm/hr)
4.001	S36	S19	10.937	0.600	94.842	94.622	0.220	49.7	375	5.81	65.8
1.007_1	S19	S20	48.683	0.600	94.547	94.079	0.468	104.0	450	8.81	53.7
5.000_1	S34	S20	45.087	0.600	94.542	94.304	0.238	189.4	225	6.34	63.2
1.008_1	S20	S21	17.043	0.600	94.079	93.463	0.616	27.7	450	8.88	53.5
1.009_1	S21	S22	10.973	0.600	93.463	93.315	0.148	74.1	450	8.96	53.2
1.010_1	S22	S23	28.365	0.600	93.315	92.945	0.370	76.7	450	9.16	52.6
1.011_1	S23	S24	13.078	0.600	92.945	92.766	0.179	73.1	450	9.25	52.4
1.012_1	S24	S25	69.678	0.600	92.692	91.720	0.972	71.7	525	9.69	51.1
1.013_1	S25	S26	79.836	0.600	91.720	91.045	0.675	118.3	525	10.34	49.4
1.014_1	S26	S27	16.262	0.600	90.820	90.663	0.157	103.6	750	10.44	49.1
1.015_1	S27	S28	50.903	0.600	90.663	90.533	0.130	391.6	750	11.12	47.5
1.016_1	S28	S29	47.420	0.600	90.533	90.467	0.066	718.5	750	11.76	46.0
1.017	S29	S30	51.049	0.600	90.467	90.397	0.070	729.3	750	12.44	44.6
1.018_1	S30	S31	59.264	0.600	90.397	90.315	0.082	722.7	750	13.23	43.1
1.019	S31	S32	46.087	0.600	90.240	90.168	0.072	640.1	825	13.82	42.0
1.020	S32	S33	14.676	0.600	90.168	90.150	0.018	815.3	825	14.00	41.7
6.000	S44	S45	72.454	0.600	94.601	92.730	1.871	38.7	450	5.72	66.3
6.001	S45	S46	59.387	0.600	92.730	91.230	1.500	39.6	450	6.29	63.5
6.002	S46	S33	18.194	0.600	91.230	90.600	0.630	28.9	450	6.48	62.6
1.021	S33	S47	34.086	0.600	90.150	90.109	0.041	831.4	900	14.91	40.2
1.022	S47	S48	41.826	0.600	90.109	90.057	0.052	804.3	900	16.48	37.9
1.023	S48	S49	35.943	0.600	90.057	89.991	0.066	544.6	900	17.83	36.1
1.024	S49	S72	48.120	0.600	89.991	89.943	0.048	1002.5	1000	19.56	34.1
1.025_1	S72	S60	48.201	0.030	89.943	89.895	0.048	1004.2	1000	20.03	33.6

Name	Vel (m/s)	Cap (l/s)	Flow (l/s)	US Depth (m)	DS Depth (m)	Σ Area (ha)	Σ Add Inflow (l/s)	Pro Depth (mm)	Pro Velocity (m/s)
4.001	2.575	284.4	45.7	1.200	1.337	0.256	0.0	101	1.905
1.007_1	1.993	316.9	302.8	1.337	1.200	2.080	0.0	355	2.253
5.000_1	0.946	37.6	12.2	1.200	1.200	0.071	0.0	87	0.844
1.008_1	3.876	616.5	329.8	1.200	1.589	2.274	0.0	234	3.938
1.009_1	2.363	375.9	345.6	1.589	1.586	2.397	0.0	342	2.664
1.010_1	2.323	369.4	359.3	1.586	1.591	2.520	0.0	361	2.628
1.011_1	2.380	378.5	375.4	1.591	1.605	2.643	0.0	369	2.692
1.012_1	2.647	573.1	383.1	1.604	1.686	2.766	0.0	315	2.827
1.013_1	2.058	445.5	386.8	1.686	1.344	2.889	0.0	380	2.303
1.014_1	2.749	1214.4	399.0	1.344	1.340	2.998	0.0	295	2.477
1.015_1	1.408	621.9	400.0	1.340	1.759	3.107	0.0	438	1.490
1.016_1	1.036	457.7	401.0	1.759	2.175	3.216	0.0	547	1.161
1.017	1.028	454.3	414.4	2.175	1.897	3.428	0.0	567	1.157
1.018_1	1.033	456.3	425.2	1.897	1.516	3.640	0.0	578	1.165
1.019	1.166	623.2	438.5	1.516	1.344	3.852	0.0	512	1.257
1.020	1.032	551.4	459.3	1.344	1.877	4.064	0.0	579	1.147
6.000	3.274	520.8	179.7	1.048	1.357	1.000	0.0	181	2.982
6.001	3.238	515.0	209.5	1.357	1.576	1.217	0.0	199	3.075
6.002	3.794	603.4	243.3	1.576	1.802	1.434	0.0	198	3.596
1.021	1.078	686.0	605.3	1.802	1.142	5.555	0.0	661	1.209
1.022	1.097	697.6	592.9	1.142	1.278	5.772	0.0	641	1.223
1.023	1.335	849.5	570.4	1.278	2.109	5.829	0.0	542	1.427
1.024	1.694	6777.3	538.8	2.009	1.057	5.829	0.0	307	0.913
1.025_1	0.703	2811.1	593.3	1.057	1.605	6.515	0.0	501	0.473

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Links

Name	US Node	DS Node	Length (m)	ks (mm) / n	US IL (m)	DS IL (m)	Fall (m)	Slope (1:X)	Dia (mm)	T of C (mins)	Rain (mm/hr)
7.000	S58	S59 Flow Control	76.047	0.600	90.520	90.375	0.145	524.5	1000	11.87	45.8
8.000	S61	S59 Flow Control	50.172	0.600	90.476	90.375	0.101	496.8	1000	11.20	47.3
7.001	S59 Flow Control	S60	41.916	0.600	90.375	90.110	0.265	158.2	300	12.43	44.6
1.025	S60	S70	12.514	0.600	89.895	89.882	0.013	962.6	1000	21.94	31.8
1.027_1	S70	S50	120.518	0.030	89.882	89.762	0.120	1004.3	1000	23.95	30.0
1.026	S50	S55	33.820	0.600	89.762	89.728	0.034	994.7	1000	22.55	31.2
1.027	S55	S51	47.942	0.600	89.728	89.680	0.048	998.8	1000	23.41	30.5
1.028	S51	S52	107.307	0.600	89.680	89.573	0.107	1002.9	1000	25.33	29.0
1.029	S52	S56	19.539	0.600	89.573	89.553	0.020	977.0	1000	25.68	28.7
1.030	S56	S57	48.443	0.600	89.553	89.505	0.048	1009.2	1000	26.55	28.1
1.031	S57	S53	49.866	0.600	89.505	89.440	0.065	767.2	1000	27.44	27.5
1.032	S53	S54 HW	23.752	0.600	89.440	89.410	0.030	791.7	1000	27.98	27.2
8.000_1	40	S74	45.777	0.600	90.443	90.327	0.116	394.6	300	15.76	38.9
7.000_1	S73	S74	52.550	0.600	90.473	90.252	0.221	237.8	300	5.88	50.0
7.001_1	S74	S71	48.442	0.600	90.252	90.097	0.155	312.5	375	16.57	37.8
7.002	S71	S72	48.342	0.600	90.097	89.943	0.154	313.9	375	17.37	36.7
11.000	S75	S76	42.747	0.600	90.453	90.311	0.142	301.0	300	5.71	50.0
12.000	S77	S76	22.761	0.600	90.387	90.311	0.076	299.5	300	5.38	50.0
11.001	S76	S68	39.488	0.600	90.311	90.179	0.132	299.2	300	6.37	50.0
11.002	S68	S69	43.916	0.600	90.179	90.033	0.146	300.8	450	7.10	50.0
11.003	S69	S70	45.015	0.600	90.033	89.882	0.151	298.1	450	7.85	50.0
13.000	S65	S66	28.011	0.600	90.301	90.183	0.118	237.4	300	5.47	50.0
13.001	S66	S67	63.330	0.600	90.108	89.906	0.202	313.5	375	6.52	50.0

Name	Vel (m/s)	Cap (l/s)	Flow (l/s)	US Depth (m)	DS Depth (m)	Σ Area (ha)	Σ Add Inflow (l/s)	Pro Depth (mm)	Pro Velocity (m/s)
7.000	2.347	9389.4	23.6	1.480	1.625	0.190	0.0	45	0.452
8.000	2.412	9649.1	24.4	1.524	1.625	0.190	0.0	45	0.464
7.001	1.247	88.2	68.9	2.325	2.090	0.570	0.0	200	1.375
1.025	1.729	6917.3	610.7	1.605	1.118	7.085	0.0	324	0.957
1.027_1	0.703	2811.0	626.9	1.118	1.738	7.709	0.0	514	0.479
1.026	1.701	6804.0	651.9	1.738	1.772	7.709	0.0	337	0.962
1.027	1.697	6789.9	663.1	1.772	1.820	8.021	0.0	340	0.964
1.028	1.694	6776.0	630.5	1.820	1.927	8.021	0.0	333	0.951
1.029	1.716	6866.0	642.6	1.927	1.947	8.261	0.0	333	0.965
1.030	1.689	6754.5	629.2	1.947	1.995	8.261	0.0	333	0.948
1.031	1.939	7754.5	615.8	1.995	2.060	8.261	0.0	307	1.045
1.032	1.180	926.9	609.0	2.060	0.190	8.261	0.0	593	1.255
8.000_1	0.785	55.5	20.9	1.675	1.470	0.198	0.0	127	0.731
7.000_1	1.015	71.7	17.3	1.073	1.545	0.128	0.0	100	0.839
7.001_1	1.019	112.6	46.5	1.470	1.220	0.454	0.0	168	0.973
7.002	1.017	112.3	57.9	1.220	1.682	0.582	0.0	191	1.024
11.000	0.901	63.7	14.1	1.441	1.282	0.104	0.0	96	0.728
12.000	0.903	63.8	14.1	1.254	1.282	0.104	0.0	95	0.728
11.001	0.904	63.9	42.3	1.282	1.393	0.312	0.0	178	0.964
11.002	1.167	185.6	56.4	1.243	1.419	0.416	0.0	169	1.027
11.003	1.172	186.4	70.5	1.419	1.668	0.520	0.0	191	1.093
13.000	1.016	71.8	14.1	1.683	1.552	0.104	0.0	90	0.795
13.001	1.018	112.4	28.2	1.552	1.209	0.208	0.0	128	0.852

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Links

Name	US Node	DS Node	Length (m)	ks (mm) / n	US IL (m)	DS IL (m)	Fall (m)	Slope (1:X)	Dia (mm)	T of C (mins)	Rain (mm/hr)
13.002	S67	S55	91.737	0.600	89.831	89.728	0.103	890.7	450	8.05	50.0
15.000	S62	S63	21.156	0.600	89.820	89.780	0.040	528.9	300	5.35	50.0
14.000	S64	S63	48.960	0.600	89.870	89.780	0.090	544.0	300	5.78	50.0
14.001	S63	S52	28.052	0.600	89.780	89.700	0.080	350.7	300	6.24	50.0

Name	Vel (m/s)	Cap (l/s)	Flow (l/s)	US Depth (m)	DS Depth (m)	Σ Area (ha)	Σ Add Inflow (l/s)	Pro Depth (mm)	Pro Velocity (m/s)
13.002	0.673	107.0	42.3	1.209	2.322	0.312	0.0	196	0.635
15.000	0.677	47.8	10.8	1.819	1.674	0.080	0.0	97	0.550
14.000	0.667	47.1	10.8	1.193	1.674	0.080	0.0	98	0.544
14.001	0.834	58.9	32.5	1.674	2.500	0.240	0.0	159	0.853

Pipeline Schedule

Link	Length (m)	Slope (1:X)	Dia (mm)	Link Type	US CL (m)	US IL (m)	US Depth (m)	DS CL (m)	DS IL (m)	DS Depth (m)
1.005	58.306	313.5	375	Circular_Default Sewer Type	92.424	91.084	0.965	92.809	90.898	1.536
1.006	48.494	418.1	450	Circular_Default Sewer Type	92.809	90.823	1.536	92.472	90.707	1.315
1.007	17.478	416.1	450	Circular_Default Sewer Type	92.472	90.707	1.315	92.376	90.665	1.261
2.000	9.176	316.4	375	Circular_Default Sewer Type	92.415	90.769	1.271	92.376	90.740	1.261
1.008	25.931	454.9	525	Circular_Default Sewer Type	92.376	90.590	1.261	92.589	90.533	1.531
1.009	81.447	455.0	525	Circular_Default Sewer Type	92.589	90.533	1.531	92.316	90.354	1.437
1.010	23.231	580.8	600	Circular_Default Sewer Type	92.316	90.279	1.437	92.406	90.239	1.567
1.211	23.334	583.4	600	Circular_Default Sewer Type	92.406	90.239	1.567	92.233	90.199	1.434
1.011	16.831	580.4	600	Circular_Default Sewer Type	92.233	90.199	1.434	92.089	90.170	1.319
5.001	9.877	240.9	300	Circular_Default Sewer Type	92.277	90.540	1.437	92.233	90.499	1.434
1.212	72.235	587.3	600	Circular_Default Sewer Type	92.089	90.170	1.319	92.229	90.047	1.582
1.012	30.700	579.2	600	Circular_Default Sewer Type	92.229	90.047	1.582	92.089	89.994	1.495
1.013	18.273	480.9	600	Circular_Default Sewer Type	92.089	89.994	1.495	91.943	89.956	1.387
1.014	45.156	716.8	750	Circular_Default Sewer Type	91.943	89.806	1.387	91.945	89.743	1.452
5.000	33.765	229.7	300	Circular_Default Sewer Type	92.151	90.330	1.521	91.945	90.183	1.462
1.015	16.393	224.6	900	Circular_Default Sewer Type	91.945	89.593	1.452	90.525	89.520	0.105

Link	US Node	Dia (mm)	Node Type	MH Type	DS Node	Dia (mm)	Node Type	MH Type
1.005	6	1350	Manhole	Adoptable	7	1350	Manhole	Adoptable
1.006	7	1350	Manhole	Adoptable	8	1350	Manhole	Adoptable
1.007	8	1350	Manhole	Adoptable	9	1350	Manhole	Adoptable
2.000	30	1350	Manhole	Adoptable	9	1350	Manhole	Adoptable
1.008	9	1350	Manhole	Adoptable	10	1500	Manhole	Adoptable
1.009	10	1500	Manhole	Adoptable	11	1500	Manhole	Adoptable
1.010	11	1500	Manhole	Adoptable	S79	1500	Manhole	Adoptable
1.211	S79	1500	Manhole	Adoptable	12	1500	Manhole	Adoptable
1.011	12	1500	Manhole	Adoptable	S78	1500	Manhole	Adoptable
5.001	50	1200	Manhole	Adoptable	12	1500	Manhole	Adoptable
1.212	S78	1500	Manhole	Adoptable	13	1500	Manhole	Adoptable
1.012	13	1500	Manhole	Adoptable	14	1500	Manhole	Adoptable
1.013	14	1500	Manhole	Adoptable	15	1800	Manhole	Adoptable
1.014	15	1800	Manhole	Adoptable	16	1800	Manhole	Adoptable
5.000	60	1200	Manhole	Adoptable	16	1800	Manhole	Adoptable
1.015	16	1800	Manhole	Adoptable	17	1800	Manhole	Adoptable

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Pipeline Schedule

Link	Length (m)	Slope (1:X)	Dia (mm)	Link Type	US CL (m)	US IL (m)	US Depth (m)	DS CL (m)	DS IL (m)	DS Depth (m)
1.016	74.839	1626.9	1050	Conveyance Swale	90.525	89.520	-0.045	90.600	89.474	0.076
16.014	19.740	308.4	525	Circular_Default Sewer Type	90.600	89.474	0.601	90.600	89.410	0.665
20.000	10.068	296.1	1050	Circular_Default Sewer Type	90.495	89.225	0.220	90.982	89.191	0.741
1.018	33.269	391.4	450	Circular_Default Sewer Type	90.982	89.191	1.341	90.250	89.106	0.694
20.002	26.431	1016.6	1000	Swale	90.250	89.106	0.144	90.250	89.080	0.170
20.003	22.040	1001.8	1000	Swale	90.250	89.080	0.170	90.250	89.058	0.192
20.004	38.323	403.4	450	Circular_Default Sewer Type	90.250	89.058	0.742	90.271	88.963	0.858
20.005	10.161	1016.1	1000	Conveyance Swale	90.271	88.963	0.308	90.270	88.953	0.317
20.006	9.345	1038.3	1000	Conveyance Swale	90.270	88.953	0.317	90.270	88.944	0.326
20.007	22.009	407.6	450	Circular_Default Sewer Type	90.270	88.944	0.876	89.837	88.890	0.497
1.000	33.250	226.2	225	Circular_Default Sewer Type	98.992	97.567	1.200	99.134	97.420	1.489
1.001	48.339	127.9	300	Circular_Default Sewer Type	99.134	97.420	1.414	98.542	97.042	1.200
1.002	34.198	76.3	300	Circular_Default Sewer Type	98.542	97.042	1.200	98.094	96.594	1.200
1.003	43.618	67.2	375	Circular_Default Sewer Type	98.094	96.519	1.200	97.525	95.870	1.280
1.004	39.245	71.7	375	Circular_Default Sewer Type	97.525	95.870	1.280	97.024	95.323	1.326
2.000_1	43.514	126.9	300	Circular_Default Sewer Type	99.256	97.503	1.453	98.810	97.160	1.350
2.001	11.133	74.2	300	Circular_Default Sewer Type	98.810	97.160	1.350	98.671	97.010	1.361
2.002	40.244	138.8	300	Circular_Default Sewer Type	98.671	97.010	1.361	98.372	96.720	1.352
3.000	41.873	73.1	300	Circular_Default Sewer Type	99.654	98.154	1.200	99.081	97.581	1.200
3.001	13.636	59.0	300	Circular_Default Sewer Type	99.081	97.581	1.200	98.868	97.350	1.218
3.002	36.713	58.3	300	Circular_Default Sewer Type	98.868	97.350	1.218	98.372	96.720	1.352
2.003	59.556	44.9	300	Circular_Default Sewer Type	98.372	96.720	1.352	97.024	95.395	1.329
1.005_1	27.054	72.0	450	Circular_Default Sewer Type	97.024	95.248	1.326	96.680	94.872	1.358

Link	US Node	Dia (mm)	Node Type	MH Type	DS Node	Dia (mm)	Node Type	MH Type
1.016	17	1800	Manhole	Adoptable	S93		Junction	
16.014	S93		Junction		S54 HW	1900	Manhole	Adoptable
20.000	S62 HW	1350	Manhole	Adoptable	Basin A Flow Control	1800	Manhole	Adoptable
1.018	Basin A Flow Control	1800	Manhole	Adoptable	Basin A to Swale		Junction	
20.002	Basin A to Swale		Junction		S87		Junction	
20.003	S87		Junction		S88		Junction	
20.004	S88		Junction		S89		Junction	
20.005	S89		Junction		S92		Junction	
20.006	S92		Junction		S90		Junction	
20.007	S90		Junction		Basin A Outfall	1350	Manhole	Adoptable
1.000	S12	1350	Manhole	Adoptable	S13	1200	Manhole	Adoptable
1.001	S13	1200	Manhole	Adoptable	S14	1350	Manhole	Adoptable
1.002	S14	1350	Manhole	Adoptable	S15	1350	Manhole	Adoptable
1.003	S15	1350	Manhole	Adoptable	S16	1350	Manhole	Adoptable
1.004	S16	1350	Manhole	Adoptable	S17	1350	Manhole	Adoptable
2.000_1	S41	1350	Manhole	Adoptable	S42	1350	Manhole	Adoptable
2.001	S42	1350	Manhole	Adoptable	S43	1350	Manhole	Adoptable
2.002	S43	1350	Manhole	Adoptable	S40	1200	Manhole	Adoptable
3.000	S37	1350	Manhole	Adoptable	S38	1350	Manhole	Adoptable
3.001	S38	1350	Manhole	Adoptable	S39	1350	Manhole	Adoptable
3.002	S39	1350	Manhole	Adoptable	S40	1200	Manhole	Adoptable
2.003	S40	1200	Manhole	Adoptable	S17	1350	Manhole	Adoptable
1.005_1	S17	1350	Manhole	Adoptable	S18	1350	Manhole	Adoptable

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Pipeline Schedule

Link	Length (m)	Slope (1:X)	Dia (mm)	Link Type	US CL (m)	US IL (m)	US Depth (m)	DS CL (m)	DS IL (m)	DS Depth (m)
1.006_1	26.084	104.3	450	Circular_Default Sewer Type	96.680	94.797	1.433	96.334	94.547	1.337
4.000	44.729	84.4	300	Circular_Default Sewer Type	96.947	95.447	1.200	96.417	94.917	1.200
4.001	10.937	49.7	375	Circular_Default Sewer Type	96.417	94.842	1.200	96.334	94.622	1.337
1.007_1	48.683	104.0	450	Circular_Default Sewer Type	96.334	94.547	1.337	95.729	94.079	1.200
5.000_1	45.087	189.4	225	Circular_Default Sewer Type	95.967	94.542	1.200	95.729	94.304	1.200
1.008_1	17.043	27.7	450	Circular_Default Sewer Type	95.729	94.079	1.200	95.502	93.463	1.589
1.009_1	10.973	74.1	450	Circular_Default Sewer Type	95.502	93.463	1.589	95.351	93.315	1.586
1.010_1	28.365	76.7	450	Circular_Default Sewer Type	95.351	93.315	1.586	94.986	92.945	1.591
1.011_1	13.078	73.1	450	Circular_Default Sewer Type	94.986	92.945	1.591	94.821	92.766	1.605
1.012_1	69.678	71.7	525	Circular_Default Sewer Type	94.821	92.692	1.604	93.931	91.720	1.686
1.013_1	79.836	118.3	525	Circular_Default Sewer Type	93.931	91.720	1.686	92.914	91.045	1.344
1.014_1	16.262	103.6	750	Circular_Default Sewer Type	92.914	90.820	1.344	92.753	90.663	1.340
1.015_1	50.903	391.6	750	Circular_Default Sewer Type	92.753	90.663	1.340	93.042	90.533	1.759
1.016_1	47.420	718.5	750	Circular_Default Sewer Type	93.042	90.533	1.759	93.392	90.467	2.175
1.017	51.049	729.3	750	Circular_Default Sewer Type	93.392	90.467	2.175	93.044	90.397	1.897
1.018_1	59.264	722.7	750	Circular_Default Sewer Type	93.044	90.397	1.897	92.581	90.315	1.516
1.019	46.087	640.1	825	Circular_Default Sewer Type	92.581	90.240	1.516	92.337	90.168	1.344
1.020	14.676	815.3	825	Circular_Default Sewer Type	92.337	90.168	1.344	92.852	90.150	1.877
6.000	72.454	38.7	450	Circular_Default Sewer Type	96.099	94.601	1.048	94.537	92.730	1.357
6.001	59.387	39.6	450	Circular_Default Sewer Type	94.537	92.730	1.357	93.256	91.230	1.576
6.002	18.194	28.9	450	Circular_Default Sewer Type	93.256	91.230	1.576	92.852	90.600	1.802
1.021	34.086	831.4	900	Circular_Default Sewer Type	92.852	90.150	1.802	92.151	90.109	1.142
1.022	41.826	804.3	900	Circular_Default Sewer Type	92.151	90.109	1.142	92.235	90.057	1.278
1.023	35.943	544.6	900	Circular_Default Sewer Type	92.235	90.057	1.278	93.000	89.991	2.109
1.024	48.120	1002.5	1000	Swale	93.000	89.991	2.009	92.000	89.943	1.057

Link	US Node	Dia (mm)	Node Type	MH Type	DS Node	Dia (mm)	Node Type	MH Type
1.006_1	S18	1350	Manhole	Adoptable	S19	1350	Manhole	Adoptable
4.000	S35	1350	Manhole	Adoptable	S36	1350	Manhole	Adoptable
4.001	S36	1350	Manhole	Adoptable	S19	1350	Manhole	Adoptable
1.007_1	S19	1350	Manhole	Adoptable	S20	1350	Manhole	Adoptable
5.000_1	S34	1350	Manhole	Adoptable	S20	1350	Manhole	Adoptable
1.008_1	S20	1350	Manhole	Adoptable	S21	1350	Manhole	Adoptable
1.009_1	S21	1350	Manhole	Adoptable	S22	1350	Manhole	Adoptable
1.010_1	S22	1350	Manhole	Adoptable	S23	1350	Manhole	Adoptable
1.011_1	S23	1350	Manhole	Adoptable	S24	1500	Manhole	Adoptable
1.012_1	S24	1500	Manhole	Adoptable	S25	1500	Manhole	Adoptable
1.013_1	S25	1500	Manhole	Adoptable	S26	1800	Manhole	Adoptable
1.014_1	S26	1800	Manhole	Adoptable	S27	1800	Manhole	Adoptable
1.015_1	S27	1800	Manhole	Adoptable	S28	1800	Manhole	Adoptable
1.016_1	S28	1800	Manhole	Adoptable	S29	1800	Manhole	Adoptable
1.017	S29	1800	Manhole	Adoptable	S30	1800	Manhole	Adoptable
1.018_1	S30	1800	Manhole	Adoptable	S31	1800	Manhole	Adoptable
1.019	S31	1800	Manhole	Adoptable	S32	1800	Manhole	Adoptable
1.020	S32	1800	Manhole	Adoptable	S33	1800	Manhole	Adoptable
6.000	S44	1350	Manhole	Adoptable	S45	1200	Manhole	Adoptable
6.001	S45	1200	Manhole	Adoptable	S46	1200	Manhole	Adoptable
6.002	S46	1200	Manhole	Adoptable	S33	1800	Manhole	Adoptable
1.021	S33	1800	Manhole	Adoptable	S47	1350	Manhole	Adoptable
1.022	S47	1350	Manhole	Adoptable	S48	1200	Manhole	Adoptable
1.023	S48	1200	Manhole	Adoptable	S49	1350	Manhole	Adoptable
1.024	S49	1350	Manhole	Adoptable	S72		Junction	

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Pipeline Schedule

Link	Length (m)	Slope (1:X)	Dia (mm)	Link Type	US CL (m)	US IL (m)	US Depth (m)	DS CL (m)	DS IL (m)	DS Depth (m)
1.025_1	48.201	1004.2	1000	Swale	92.000	89.943	1.057	92.500	89.895	1.605
7.000	76.047	524.5	1000	Swale	93.000	90.520	1.480	93.000	90.375	1.625
8.000	50.172	496.8	1000	Swale	93.000	90.476	1.524	93.000	90.375	1.625
7.001	41.916	158.2	300	Circular_Default Sewer Type	93.000	90.375	2.325	92.500	90.110	2.090
1.025	12.514	962.6	1000	Swale	92.500	89.895	1.605	92.000	89.882	1.118
1.027_1	120.518	1004.3	1000	Swale	92.000	89.882	1.118	92.500	89.762	1.738
1.026	33.820	994.7	1000	Swale	92.500	89.762	1.738	92.500	89.728	1.772
1.027	47.942	998.8	1000	Swale	92.500	89.728	1.772	92.500	89.680	1.820
1.028	107.307	1002.9	1000	Swale	92.500	89.680	1.820	92.500	89.573	1.927
1.029	19.539	977.0	1000	Swale	92.500	89.573	1.927	92.500	89.553	1.947
1.030	48.443	1009.2	1000	Swale	92.500	89.553	1.947	92.500	89.505	1.995
1.031	49.866	767.2	1000	Swale	92.500	89.505	1.995	92.500	89.440	2.060
1.032	23.752	791.7	1000	Circular_Default Sewer Type	92.500	89.440	2.060	90.600	89.410	0.190
8.000_1	45.777	394.6	300	Circular_Default Sewer Type	92.418	90.443	1.675	92.097	90.327	1.470
7.000_1	52.550	237.8	300	Circular_Default Sewer Type	91.846	90.473	1.073	92.097	90.252	1.545
7.001_1	48.442	312.5	375	Circular_Default Sewer Type	92.097	90.252	1.470	91.692	90.097	1.220
7.002	48.342	313.9	375	Circular_Default Sewer Type	91.692	90.097	1.220	92.000	89.943	1.682
11.000	42.747	301.0	300	Circular_Default Sewer Type	92.194	90.453	1.441	91.893	90.311	1.282
12.000	22.761	299.5	300	Circular_Default Sewer Type	91.941	90.387	1.254	91.893	90.311	1.282
11.001	39.488	299.2	300	Circular_Default Sewer Type	91.893	90.311	1.282	91.872	90.179	1.393
11.002	43.916	300.8	450	Circular_Default Sewer Type	91.872	90.179	1.243	91.902	90.033	1.419
11.003	45.015	298.1	450	Circular_Default Sewer Type	91.902	90.033	1.419	92.000	89.882	1.668
13.000	28.011	237.4	300	Circular_Default Sewer Type	92.284	90.301	1.683	92.035	90.183	1.552
13.001	63.330	313.5	375	Circular_Default Sewer Type	92.035	90.108	1.552	91.490	89.906	1.209

Link	US Node	Dia (mm)	Node Type	MH Type	DS Node	Dia (mm)	Node Type	MH Type
1.025_1	S72		Junction		S60	1350	Manhole	Adoptable
7.000	S58	1350	Manhole	Adoptable	S59 Flow Control	1350	Manhole	Adoptable
8.000	S61	1350	Manhole	Adoptable	S59 Flow Control	1350	Manhole	Adoptable
7.001	S59 Flow Control	1350	Manhole	Adoptable	S60	1350	Manhole	Adoptable
1.025	S60	1350	Manhole	Adoptable	S70		Junction	
1.027_1	S70		Junction		S50	1350	Manhole	Adoptable
1.026	S50	1350	Manhole	Adoptable	S55	1350	Manhole	Adoptable
1.027	S55	1350	Manhole	Adoptable	S51	1350	Manhole	Adoptable
1.028	S51	1350	Manhole	Adoptable	S52	1350	Manhole	Adoptable
1.029	S52	1350	Manhole	Adoptable	S56	1350	Manhole	Adoptable
1.030	S56	1350	Manhole	Adoptable	S57	1350	Manhole	Adoptable
1.031	S57	1350	Manhole	Adoptable	S53	1350	Manhole	Adoptable
1.032	S53	1350	Manhole	Adoptable	S54 HW	1900	Manhole	Adoptable
8.000_1	40	1200	Manhole	Adoptable	S74	1350	Manhole	Adoptable
7.000_1	S73	1350	Manhole	Adoptable	S74	1350	Manhole	Adoptable
7.001_1	S74	1350	Manhole	Adoptable	S71	1350	Manhole	Adoptable
7.002	S71	1350	Manhole	Adoptable	S72		Junction	
11.000	S75	1350	Manhole	Adoptable	S76	1350	Manhole	Adoptable
12.000	S77	1350	Manhole	Adoptable	S76	1350	Manhole	Adoptable
11.001	S76	1350	Manhole	Adoptable	S68	1350	Manhole	Adoptable
11.002	S68	1350	Manhole	Adoptable	S69	1350	Manhole	Adoptable
11.003	S69	1350	Manhole	Adoptable	S70		Junction	
13.000	S65	1350	Manhole	Adoptable	S66	1350	Manhole	Adoptable
13.001	S66	1350	Manhole	Adoptable	S67	1350	Manhole	Adoptable

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Pipeline Schedule

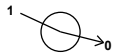
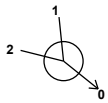
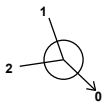
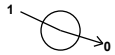
Link	Length (m)	Slope (1:X)	Dia (mm)	Link Type	US CL (m)	US IL (m)	US Depth (m)	DS CL (m)	DS IL (m)	DS Depth (m)
13.002	91.737	890.7	450	Circular_Default Sewer Type	91.490	89.831	1.209	92.500	89.728	2.322
15.000	21.156	528.9	300	Circular_Default Sewer Type	91.939	89.820	1.819	91.754	89.780	1.674
14.000	48.960	544.0	300	Circular_Default Sewer Type	91.363	89.870	1.193	91.754	89.780	1.674
14.001	28.052	350.7	300	Circular_Default Sewer Type	91.754	89.780	1.674	92.500	89.700	2.500

Link	US Node	Dia (mm)	Node Type	MH Type	DS Node	Dia (mm)	Node Type	MH Type
13.002	S67	1350	Manhole	Adoptable	S55	1350	Manhole	Adoptable
15.000	S62	1350	Manhole	Adoptable	S63	1350	Manhole	Adoptable
14.000	S64	1350	Manhole	Adoptable	S63	1350	Manhole	Adoptable
14.001	S63	1350	Manhole	Adoptable	S52	1350	Manhole	Adoptable

Manhole Schedule

Node	Easting (m)	Northing (m)	CL (m)	Depth (m)	Dia (mm)	Connections		Link	IL (m)	Dia (mm)
6	419574.518	187451.974	92.424	1.340	1350		0	1.005	91.084	375
7	419618.192	187490.602	92.809	1.986	1350		1	1.005	90.898	375
							0	1.006	90.823	450
8	419654.412	187522.847	92.472	1.765	1350		1	1.006	90.707	450
							0	1.007	90.707	450
30	419670.494	187538.459	92.415	1.646	1350		0	2.000	90.769	375
9	419670.661	187529.285	92.376	1.786	1350		1	2.000	90.740	375
							2	1.007	90.665	450
							0	1.008	90.590	525
10	419694.470	187519.011	92.589	2.056	1500		1	1.008	90.533	525
							0	1.009	90.533	525
40	419744.787	187435.344	92.418	1.975	1200		0	8.000_1	90.443	300
11	419751.281	187460.113	92.316	2.037	1500		1	1.009	90.354	525
							0	1.010	90.279	600
50	419793.986	187454.517	92.277	1.737	1200		0	5.001	90.540	300

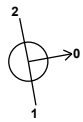
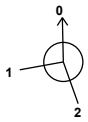
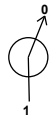
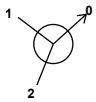
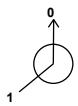
Manhole Schedule

Node	Easting (m)	Northing (m)	CL (m)	Depth (m)	Dia (mm)	Connections		Link	IL (m)	Dia (mm)
S79	419772.361	187450.351	92.406	2.167	1500		1	1.010	90.239	600
							0	1.211	90.239	600
12	419794.998	187444.692	92.233	2.034	1500		1	5.001	90.499	300
							2	1.211	90.199	600
							0	1.011	90.199	600
S78	419808.120	187434.152	92.089	1.919	1500		1	1.011	90.170	600
							0	1.212	90.170	600
13	419852.097	187376.846	92.229	2.182	1500		1	1.212	90.047	600
							0	1.012	90.047	600
14	419873.118	187354.472	92.089	2.095	1500		1	1.012	89.994	600
							0	1.013	89.994	600
15	419890.913	187350.318	91.943	2.137	1800		1	1.013	89.956	600
							0	1.014	89.806	750
60	419924.454	187388.859	92.151	1.821	1200		0	5.000	90.330	300
16	419935.575	187356.978	91.945	2.352	1800		1	5.000	90.183	300
							2	1.014	89.743	750
							0	1.015	89.593	900
17	419954.275	187339.070	90.525	1.005	1800		1	1.015	89.520	900
							0	1.016	89.520	1050
18	419999.287	187314.408	90.800		1800					
Basin A Flow Control	420139.953	187253.093	90.982	1.791	1800		1	20.000	89.191	1050
							0	1.018	89.191	450
Basin A Outfall	420222.346	187274.600	89.837	0.947	1350		1	20.007	88.890	450
Basin A to Swale	420173.163	187242.871	90.250	1.144			1	1.018	89.106	450
							0	20.002	89.106	1000

Manhole Schedule

Node	Easting (m)	Northing (m)	CL (m)	Depth (m)	Dia (mm)	Connections		Link	IL (m)	Dia (mm)
S87	420196.762	187230.967	90.250	1.170			1	20.002	89.080	1000
							0	20.003	89.080	1000
S88	420218.697	187228.818	90.250	1.192			1	20.003	89.058	1000
							0	20.004	89.058	450
S89	420247.892	187253.643	90.271	1.308			1	20.004	88.963	450
							0	20.005	88.963	1000
S92	420244.486	187263.216	90.270	1.317			1	20.005	88.953	1000
							0	20.006	88.953	1000
S90	420244.260	187272.558	90.270	1.326			1	20.006	88.944	1000
							0	20.007	88.944	450
S12	419219.225	187158.445	98.992	1.425	1350		0	1.000	97.567	225
S13	419212.887	187191.085	99.134	1.714	1200		1	1.000	97.420	225
							0	1.001	97.420	300
S14	419212.734	187239.424	98.542	1.500	1350		1	1.001	97.042	300
							0	1.002	97.042	300
S15	419199.040	187270.760	98.094	1.575	1350		1	1.002	96.594	300
							0	1.003	96.519	375
S16	419176.176	187307.905	97.525	1.655	1350		1	1.003	95.870	375
							0	1.004	95.870	375
S41	419051.014	187353.832	99.256	1.753	1350		0	2.000_1	97.503	300
S42	419086.623	187378.839	98.810	1.650	1350		1	2.000_1	97.160	300
							0	2.001	97.160	300
S43	419096.534	187373.769	98.671	1.661	1350		1	2.001	97.010	300
							0	2.002	97.010	300

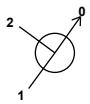
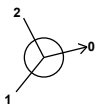
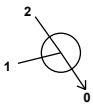
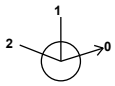
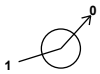
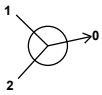
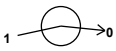
Manhole Schedule

Node	Easting (m)	Northing (m)	CL (m)	Depth (m)	Dia (mm)	Connections		Link	IL (m)	Dia (mm)
S37	419072.025	187262.568	99.654	1.500	1350		0	3.000	98.154	300
S38	419107.438	187284.912	99.081	1.500	1350		1	3.000	97.581	300
							0	3.001	97.581	300
S39	419110.745	187298.140	98.868	1.518	1350		1	3.001	97.350	300
							0	3.002	97.350	300
S40	419104.414	187334.303	98.372	1.652	1200		1	3.002	96.720	300
							2	2.002	96.720	300
							0	2.003	96.720	300
S17	419163.023	187344.880	97.024	1.776	1350		1	2.003	95.395	300
							2	1.004	95.323	375
							0	1.005_1	95.248	450
S18	419162.434	187371.928	96.680	1.883	1350		1	1.005_1	94.872	450
							0	1.006_1	94.797	450
S35	419136.639	187438.587	96.947	1.500	1350		0	4.000	95.447	300
S36	419163.393	187402.742	96.417	1.575	1350		1	4.000	94.917	300
							0	4.001	94.842	375
S19	419172.119	187396.148	96.334	1.787	1350		1	4.001	94.622	375
							2	1.006_1	94.547	450
							0	1.007_1	94.547	450
S34	419180.260	187464.095	95.967	1.425	1350		0	5.000_1	94.542	225
S20	419208.265	187428.759	95.729	1.650	1350		1	5.000_1	94.304	225
							2	1.007_1	94.079	450
							0	1.008_1	94.079	450
S21	419221.439	187439.571	95.502	2.039	1350		1	1.008_1	93.463	450
							0	1.009_1	93.463	450
S22	419221.523	187450.544	95.351	2.036	1350		1	1.009_1	93.315	450
							0	1.010_1	93.315	450

Manhole Schedule

Node	Easting (m)	Northing (m)	CL (m)	Depth (m)	Dia (mm)	Connections		Link	IL (m)	Dia (mm)
S23	419207.703	187475.315	94.986	2.041	1350		1	1.010_1	92.945	450
							0	1.011_1	92.945	450
S24	419212.033	187487.655	94.821	2.129	1500		1	1.011_1	92.766	450
							0	1.012_1	92.692	525
S25	419269.702	187526.762	93.931	2.211	1500		1	1.012_1	91.720	525
							0	1.013_1	91.720	525
S26	419336.421	187570.606	92.914	2.094	1800		1	1.013_1	91.045	525
							0	1.014_1	90.820	750
S27	419352.227	187566.780	92.753	2.090	1800		1	1.014_1	90.663	750
							0	1.015_1	90.663	750
S28	419392.176	187535.233	93.042	2.509	1800		1	1.015_1	90.533	750
							0	1.016_1	90.533	750
S29	419416.168	187494.330	93.392	2.925	1800		1	1.016_1	90.467	750
							0	1.017	90.467	750
S30	419451.502	187457.486	93.044	2.647	1800		1	1.017	90.397	750
							0	1.018_1	90.397	750
S31	419477.685	187404.320	92.581	2.341	1800		1	1.018_1	90.315	750
							0	1.019	90.240	825
S32	419512.697	187374.351	92.337	2.169	1800		1	1.019	90.168	825
							0	1.020	90.168	825
S44	419413.187	187266.592	96.099	1.498	1350					
							0	6.000	94.601	450
S45	419472.386	187308.367	94.537	1.807	1200		1	6.000	92.730	450
							0	6.001	92.730	450
S46	419513.739	187350.990	93.256	2.026	1200		1	6.001	91.230	450
							0	6.002	91.230	450

Manhole Schedule

Node	Easting (m)	Northing (m)	CL (m)	Depth (m)	Dia (mm)	Connections		Link	IL (m)	Dia (mm)
S33	419524.515	187365.649	92.852	2.702	1800		1	6.002	90.600	450
							2	1.020	90.150	825
							0	1.021	90.150	900
S47	419544.212	187393.468	92.151	2.042	1350		1	1.021	90.109	900
							0	1.022	90.109	900
S48	419560.216	187432.111	92.235	2.178	1200		1	1.022	90.057	900
							0	1.023	90.057	900
S49	419591.449	187414.323	93.000	3.009	1350		1	1.023	89.991	900
							0	1.024	89.991	1000
S58	419569.462	187391.824	93.000	2.480	1350		0	7.000	90.520	1000
S61	419572.793	187285.223	93.000	2.524	1350		0	8.000	90.476	1000
S59 Flow Control	419604.332	187324.243	93.000	2.625	1350		1	8.000	90.375	1000
							2	7.000	90.375	1000
							0	7.001	90.375	300
S60	419645.030	187334.273	92.500	2.605	1350		1	7.001	90.110	300
							2	1.025_1	89.895	1000
							0	1.025	89.895	1000
S50	419719.115	187223.779	92.500	2.738	1350		1	1.027_1	89.762	1000
							0	1.026	89.762	1000
S55	419750.577	187211.371	92.500	2.772	1350		1	13.002	89.728	450
							2	1.026	89.728	1000
							0	1.027	89.728	1000
S51	419796.344	187225.646	92.500	2.820	1350		1	1.027	89.680	1000
							0	1.028	89.680	1000
S52	419868.460	187305.108	92.500	2.927	1350		1	14.001	89.700	300
							2	1.028	89.573	1000
							0	1.029	89.573	1000
S56	419887.557	187309.239	92.500	2.947	1350		1	1.029	89.553	1000
							0	1.030	89.553	1000

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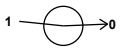
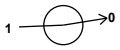
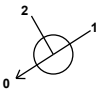
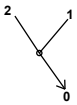
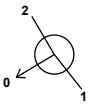
Manhole Schedule

Node	Easting (m)	Northing (m)	CL (m)	Depth (m)	Dia (mm)	Connections		Link	IL (m)	Dia (mm)
S57	419935.749	187304.315	92.500	2.995	1350		1	1.030	89.505	1000
							0	1.031	89.505	1000
S53	419993.281	187306.638	92.500	3.060	1350		1	1.031	89.440	1000
							0	1.032	89.440	1000
S54 HW	420013.487	187309.882	90.600	1.372	1900		1	16.014	89.410	525
							2	1.032	89.410	1000
S62 HW	420130.841	187257.374	90.495	1.270	1350		0	20.000	89.225	1050
S93	419995.735	187318.516	90.600	1.126			1	1.016	89.474	1050
							0	16.014	89.474	525
S73	419680.300	187456.316	91.846	1.373	1350		0	7.000_1	90.473	300
S74	419706.258	187410.625	92.097	1.845	1350		1	8.000_1	90.327	300
							2	7.000_1	90.252	300
							0	7.001_1	90.252	375
S71	419665.486	187384.467	91.692	1.595	1350		1	7.001_1	90.097	375
							0	7.002	90.097	375
S72	419634.976	187348.394	92.000	2.057			1	7.002	89.943	375
							2	1.024	89.943	1000
							0	1.025_1	89.943	1000
S75	419745.058	187416.105	92.194	1.741	1350		0	11.000	90.453	300
S76	419766.732	187379.260	91.893	1.582	1350		1	12.000	90.311	300
							2	11.000	90.311	300
							0	11.001	90.311	300
S77	419780.835	187361.395	91.941	1.554	1350		0	12.000	90.387	300
S68	419732.848	187358.983	91.872	1.693	1350		1	11.001	90.179	300
							0	11.002	90.179	450

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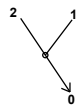
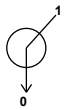
Manhole Schedule

Node	Easting (m)	Northing (m)	CL (m)	Depth (m)	Dia (mm)	Connections		Link	IL (m)	Dia (mm)
S69	419695.286	187336.230	91.902	1.869	1350		1	11.002	90.033	450
							0	11.003	90.033	450
S70	419667.902	187300.161	92.000	2.118			1	11.003	89.882	450
							2	1.025	89.882	1000
							0	1.027_1	89.882	1000
S65	419811.543	187371.010	92.284	1.983	1350		0	13.000	90.301	300
S66	419792.856	187350.144	92.035	1.927	1350		1	13.000	90.183	300
							0	13.001	90.108	375
S67	419750.450	187303.108	91.490	1.659	1350		1	13.001	89.906	375
							0	13.002	89.831	450
S62	419862.037	187340.843	91.939	2.119	1350		0	15.000	89.820	300
S63	419848.374	187324.690	91.754	1.974	1350		1	15.000	89.780	300
							2	14.000	89.780	300
							0	14.001	89.780	300
S64	419815.658	187288.265	91.363	1.493	1350		0	14.000	89.870	300

Simulation Settings

Rainfall Methodology	FSR	Analysis Speed	Normal
FSR Region	England and Wales	Skip Steady State	✓
M5-60 (mm)	20.000	Drain Down Time (mins)	1440
Ratio-R	0.400	Additional Storage (m³/ha)	0.0
Summer CV	0.750	Check Discharge Rate(s)	x
Winter CV	0.840	Check Discharge Volume	x

Storm Durations

15	30	60	120	180	240

Return Period (years)	Climate Change (CC %)	Additional Area (A %)	Additional Flow (Q %)
2	0	0	0
30	0	0	0
100	0	0	0
100	40	0	0

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Node Basin A Flow Control Online Hydro-Brake® Control

Flap Valve	x	Objective	(HE) Minimise upstream storage
Replaces Downstream Link	✓	Sump Available	✓
Invert Level (m)	89.191	Product Number	CTL-SHE-0262-3890-1200-3890
Design Depth (m)	1.200	Min Outlet Diameter (m)	0.300
Design Flow (l/s)	38.9	Min Node Diameter (mm)	1800

Node S59 Flow Control Online Hydro-Brake® Control

Flap Valve	x	Objective	(HE) Minimise upstream storage
Replaces Downstream Link	✓	Sump Available	✓
Invert Level (m)	90.375	Product Number	CTL-SHE-0077-2600-1000-2600
Design Depth (m)	1.000	Min Outlet Diameter (m)	0.100
Design Flow (l/s)	2.6	Min Node Diameter (mm)	1200

Node S62 HW Flow through Pond Storage Structure

Base Inf Coefficient (m/hr)	0.00000	Porosity	1.00	Main Channel Length (m)	123.000
Side Inf Coefficient (m/hr)	0.00000	Invert Level (m)	89.225	Main Channel Slope (1:X)	40000.0
Safety Factor	2.0	Time to half empty (mins)		Main Channel n	0.015

Inlets

S54 HW

Depth (m)	Area (m²)	Inf Area (m²)	Depth (m)	Area (m²)	Inf Area (m²)	Depth (m)	Area (m²)	Inf Area (m²)	Depth (m)	Area (m²)	Inf Area (m²)
0.000	4231.0	0.0	0.400	4582.1	0.0	0.800	4943.6	0.0	1.200	5314.2	0.0
0.100	4316.8	0.0	0.500	4671.6	0.0	0.900	5035.4	0.0	1.295	5403.2	0.0
0.200	4404.7	0.0	0.600	4761.7	0.0	1.000	5127.7	0.0			
0.300	4493.1	0.0	0.700	4852.4	0.0	1.100	5220.7	0.0			

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Results for 2 year Critical Storm Duration. Lowest mass balance: 99.78%

Node Event	US Node	Peak (mins)	Level (m)	Depth (m)	Inflow (l/s)	Node Vol (m³)	Flood (m³)	Status
15 minute summer	6	1	91.084	0.000	0.0	0.0000	0.0000	OK
15 minute winter	7	14	90.958	0.135	9.6	0.1932	0.0000	OK
15 minute winter	8	14	90.956	0.249	90.7	0.3568	0.0000	OK
15 minute winter	30	15	90.898	0.129	21.5	0.1847	0.0000	OK
15 minute winter	9	14	90.877	0.287	114.0	0.4104	0.0000	OK
15 minute winter	10	15	90.809	0.276	118.7	0.4870	0.0000	OK
30 minute winter	40	23	90.568	0.125	19.1	0.1410	0.0000	OK
15 minute winter	11	16	90.622	0.343	136.5	0.6057	0.0000	OK
30 minute winter	50	23	90.674	0.134	25.5	0.1515	0.0000	OK
15 minute winter	S79	16	90.588	0.349	134.7	0.6160	0.0000	OK
15 minute winter	12	16	90.555	0.356	159.6	0.6295	0.0000	OK
15 minute winter	S78	16	90.507	0.337	159.4	0.5958	0.0000	OK
15 minute winter	13	16	90.398	0.351	171.6	0.6210	0.0000	OK
15 minute winter	14	16	90.316	0.322	176.2	0.5691	0.0000	OK
15 minute winter	15	16	90.133	0.327	190.5	0.8311	0.0000	OK
15 minute winter	60	10	90.514	0.184	48.3	0.2084	0.0000	OK
30 minute winter	16	24	89.876	0.283	206.8	0.7204	0.0000	OK
30 minute winter	17	26	89.815	0.295	206.2	0.7502	0.0000	OK
	18							
240 minute winter	Basin A Flow Control	260	89.598	0.407	37.7	1.0366	0.0000	OK
240 minute winter	Basin A Outfall	1444	89.290	0.400	37.7	0.0000	0.0000	OK
240 minute winter	Basin A to Swale	268	89.315	0.209	37.7	0.0000	0.0000	OK
240 minute winter	S87	268	89.312	0.232	37.7	0.0000	0.0000	OK
240 minute winter	S88	268	89.311	0.253	37.7	0.0000	0.0000	OK
240 minute winter	S89	1452	89.290	0.327	37.7	0.0000	0.0000	OK

Link Event (Upstream Depth)	US Node	Link	DS Node	Outflow (l/s)	Velocity (m/s)	Flow/Cap	Link Vol (m³)	Discharge Vol (m³)
15 minute summer	6	1.005	7	0.0	0.000	0.000	0.2106	
15 minute winter	7	1.006	8	12.2	0.275	0.078	3.1538	
15 minute winter	8	1.007	9	90.1	1.111	0.572	1.4282	
15 minute winter	30	2.000	9	21.5	0.777	0.192	0.3200	
15 minute winter	9	1.008	10	113.7	1.006	0.504	3.0498	
15 minute winter	10	1.009	11	118.4	1.092	0.524	9.1680	
30 minute winter	40	8.000_1	S74	19.0	0.769	0.343	1.1354	
15 minute winter	11	1.010	S79	134.7	0.835	0.475	3.9046	
30 minute winter	50	5.001	12	25.5	0.890	0.358	0.2829	
15 minute winter	S79	1.211	12	133.4	0.795	0.472	4.0148	
15 minute winter	12	1.011	S78	159.4	0.963	0.562	2.8390	
15 minute winter	S78	1.212	13	158.4	0.950	0.562	12.0820	
15 minute winter	13	1.012	14	170.5	1.048	0.600	4.9969	
15 minute winter	14	1.013	15	176.0	1.269	0.564	2.5336	
15 minute winter	15	1.014	16	190.2	1.184	0.415	7.2623	
15 minute winter	60	5.000	16	47.0	1.102	0.644	1.4504	
30 minute winter	16	1.015	17	206.2	1.278	0.155	2.8566	
30 minute winter	17	1.016	S93	196.9	0.517	0.039	33.2304	
240 minute winter	Basin A Flow Control	Hydro-Brake®	Basin A to Swale	37.7				
240 minute winter	Basin A to Swale	20.002	S87	37.7	0.114	0.013	9.7095	
240 minute winter	S87	20.003	S88	37.7	0.090	0.013	9.2381	
240 minute winter	S88	20.004	S89	37.7	0.426	0.612	3.9459	
240 minute winter	S89	20.005	S92	37.7	0.129	0.016	5.0511	

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Results for 2 year Critical Storm Duration. Lowest mass balance: 99.78%

Node Event	US Node	Peak (mins)	Level (m)	Depth (m)	Inflow (l/s)	Node Vol (m³)	Flood (m³)	Status
240 minute winter	S92	1452	89.290	0.337	37.7	0.0000	0.0000	OK
240 minute winter	S90	1452	89.290	0.346	37.7	0.0000	0.0000	OK
30 minute winter	S12	23	97.624	0.057	5.0	0.0822	0.0000	OK
30 minute winter	S13	23	97.484	0.064	10.0	0.0724	0.0000	OK
30 minute winter	S14	24	97.112	0.070	15.0	0.1008	0.0000	OK
30 minute winter	S15	23	96.610	0.091	32.2	0.1299	0.0000	OK
30 minute winter	S16	23	95.986	0.116	49.4	0.1663	0.0000	OK
30 minute winter	S41	23	97.574	0.071	12.3	0.1015	0.0000	OK
30 minute winter	S42	23	97.257	0.097	24.6	0.1389	0.0000	OK
30 minute winter	S43	23	97.143	0.133	36.9	0.1903	0.0000	OK
30 minute winter	S37	23	98.216	0.062	12.3	0.0885	0.0000	OK
30 minute winter	S38	23	97.668	0.087	24.6	0.1251	0.0000	OK
30 minute winter	S39	23	97.452	0.102	36.9	0.1459	0.0000	OK
30 minute winter	S40	23	96.877	0.157	86.1	0.1780	0.0000	OK
30 minute winter	S17	23	95.463	0.215	152.5	0.3070	0.0000	OK
30 minute winter	S18	23	95.059	0.262	169.6	0.3751	0.0000	OK
30 minute winter	S35	23	95.512	0.065	12.3	0.0933	0.0000	OK
30 minute winter	S36	23	94.916	0.074	24.6	0.1055	0.0000	OK
30 minute winter	S19	24	94.824	0.277	199.1	0.3963	0.0000	OK
30 minute winter	S34	24	94.607	0.065	6.8	0.0934	0.0000	OK
30 minute winter	S20	24	94.282	0.203	217.8	0.2904	0.0000	OK
30 minute winter	S21	24	93.783	0.320	229.6	0.4582	0.0000	OK
30 minute winter	S22	24	93.614	0.299	241.3	0.4277	0.0000	OK
30 minute winter	S23	24	93.267	0.322	253.0	0.4606	0.0000	OK

Link Event (Upstream Depth)	US Node	Link	DS Node	Outflow (l/s)	Velocity (m/s)	Flow/Cap	Link Vol (m³)	Discharge Vol (m³)
240 minute winter	S92	20.006	S90	37.7	0.122	0.016	4.8689	
240 minute winter	S90	20.007	Basin A Outfall	37.7	0.579	0.616	3.0782	1660.1
30 minute winter	S12	1.000	S13	5.0	0.579	0.145	0.2871	
30 minute winter	S13	1.001	S14	10.0	0.847	0.102	0.5694	
30 minute winter	S14	1.002	S15	14.9	1.206	0.117	0.4228	
30 minute winter	S15	1.003	S16	32.1	1.301	0.132	1.0805	
30 minute winter	S16	1.004	S17	49.4	1.511	0.209	1.3017	
30 minute winter	S41	2.000_1	S42	12.3	0.763	0.125	0.7052	
30 minute winter	S42	2.001	S43	24.6	0.990	0.190	0.2775	
30 minute winter	S43	2.002	S40	36.9	1.092	0.392	1.3594	
30 minute winter	S37	3.000	S38	12.3	0.902	0.095	0.5753	
30 minute winter	S38	3.001	S39	24.6	1.292	0.170	0.2600	
30 minute winter	S39	3.002	S40	36.9	1.268	0.253	1.0740	
30 minute winter	S40	2.003	S17	85.9	2.353	0.517	2.1755	
30 minute winter	S17	1.005_1	S18	152.5	2.178	0.400	1.8945	
30 minute winter	S18	1.006_1	S19	169.7	1.713	0.536	2.5839	
30 minute winter	S35	4.000	S36	12.3	1.106	0.102	0.4973	
30 minute winter	S36	4.001	S19	24.6	1.022	0.086	0.4138	
30 minute winter	S19	1.007_1	S20	199.3	2.320	0.629	4.1788	
30 minute winter	S34	5.000_1	S20	6.8	0.721	0.181	0.4253	
30 minute winter	S20	1.008_1	S21	217.9	2.280	0.353	1.6191	
30 minute winter	S21	1.009_1	S22	229.6	1.975	0.611	1.2753	
30 minute winter	S22	1.010_1	S23	241.3	2.069	0.653	3.3063	
30 minute winter	S23	1.011_1	S24	252.9	2.318	0.668	1.4238	

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Results for 2 year Critical Storm Duration. Lowest mass balance: 99.78%

Node Event	US Node	Peak (mins)	Level (m)	Depth (m)	Inflow (l/s)	Node Vol (m³)	Flood (m³)	Status
30 minute winter	S24	24	92.945	0.253	264.6	0.4464	0.0000	OK
30 minute winter	S25	25	92.030	0.310	276.3	0.5470	0.0000	OK
30 minute winter	S26	25	91.145	0.325	286.1	0.8265	0.0000	OK
30 minute winter	S27	26	91.096	0.433	296.0	1.1026	0.0000	OK
30 minute winter	S28	27	91.040	0.507	303.9	1.2907	0.0000	OK
30 minute winter	S29	27	90.981	0.514	320.1	1.3082	0.0000	OK
30 minute winter	S30	27	90.910	0.513	336.3	1.3061	0.0000	OK
30 minute winter	S31	27	90.817	0.577	353.9	1.4690	0.0000	OK
30 minute winter	S32	27	90.774	0.606	372.1	1.5422	0.0000	OK
30 minute winter	S44	23	94.730	0.129	96.3	0.1849	0.0000	OK
30 minute winter	S45	22	92.878	0.148	120.1	0.1669	0.0000	OK
30 minute winter	S46	22	91.395	0.165	144.5	0.1871	0.0000	OK
30 minute winter	S33	26	90.742	0.592	491.3	1.5059	0.0000	OK
30 minute winter	S47	27	90.666	0.557	498.0	0.7976	0.0000	OK
30 minute winter	S48	27	90.555	0.498	501.5	0.5636	0.0000	OK
30 minute winter	S49	29	90.420	0.429	500.5	0.6137	0.0000	OK
240 minute winter	S58	236	90.768	0.248	6.6	0.3556	0.0000	OK
240 minute winter	S61	232	90.769	0.293	6.6	0.4192	0.0000	OK
240 minute winter	S59 Flow Control	232	90.769	0.394	15.0	0.5631	0.0000	SURCHARGED
30 minute winter	S60	30	90.364	0.469	515.3	0.6716	0.0000	OK
30 minute winter	S50	32	90.090	0.328	529.8	0.4687	0.0000	OK
30 minute winter	S55	33	90.049	0.321	538.8	0.4595	0.0000	OK
30 minute winter	S51	34	89.995	0.315	535.7	0.4507	0.0000	OK
30 minute winter	S52	37	89.949	0.376	531.7	0.5375	0.0000	OK
30 minute winter	S56	37	89.935	0.382	506.1	0.5465	0.0000	OK

Link Event (Upstream Depth)	US Node	Link	DS Node	Outflow (l/s)	Velocity (m/s)	Flow/Cap	Link Vol (m³)	Discharge Vol (m³)
30 minute winter	S24	1.012_1	S25	264.6	2.252	0.462	8.1825	
30 minute winter	S25	1.013_1	S26	276.0	2.160	0.619	10.2004	
30 minute winter	S26	1.014_1	S27	285.9	1.327	0.235	3.6246	
30 minute winter	S27	1.015_1	S28	293.8	1.058	0.472	14.7534	
30 minute winter	S28	1.016_1	S29	301.2	0.964	0.658	15.1394	
30 minute winter	S29	1.017	S30	318.3	0.996	0.701	16.4065	
30 minute winter	S30	1.018_1	S31	336.0	1.064	0.736	18.8009	
30 minute winter	S31	1.019	S32	355.2	0.877	0.570	18.8507	
30 minute winter	S32	1.020	S33	374.0	0.913	0.678	6.0819	
30 minute winter	S44	6.000	S45	96.3	2.341	0.185	2.9943	
30 minute winter	S45	6.001	S46	120.1	2.457	0.233	2.9094	
30 minute winter	S46	6.002	S33	144.5	2.950	0.239	0.8917	
30 minute winter	S33	1.021	S47	490.5	1.149	0.715	14.5593	
30 minute winter	S47	1.022	S48	496.5	1.292	0.712	16.1562	
30 minute winter	S48	1.023	S49	500.5	1.550	0.589	11.7997	
30 minute winter	S49	1.024	S72	493.4	0.480	0.073	50.9254	
240 minute winter	S58	7.000	S59 Flow Control	5.0	0.077	0.001	49.0976	
240 minute winter	S61	8.000	S59 Flow Control	5.4	0.093	0.001	35.3360	
240 minute winter	S59 Flow Control	Hydro-Brake®	S60	2.6				
30 minute winter	S60	1.025	S70	512.9	0.447	0.074	14.3730	
30 minute winter	S50	1.026	S55	523.1	0.822	0.077	21.6333	
30 minute winter	S55	1.027	S51	535.7	0.880	0.079	29.7256	
30 minute winter	S51	1.028	S52	526.5	0.834	0.078	75.1961	
30 minute winter	S52	1.029	S56	506.1	0.794	0.074	15.8119	
30 minute winter	S56	1.030	S57	494.2	0.787	0.073	42.4460	

Results for 2 year Critical Storm Duration. Lowest mass balance: 99.78%

Node Event	US Node	Peak (mins)	Level (m)	Depth (m)	Inflow (l/s)	Node Vol (m³)	Flood (m³)	Status
30 minute winter	S57	38	89.921	0.416	494.2	0.5947	0.0000	OK
30 minute winter	S53	38	89.912	0.472	481.8	0.6760	0.0000	OK
240 minute winter	S54 HW	260	89.599	0.371	324.9	1.0509	0.0000	OK
240 minute winter	S62 HW	260	89.599	0.374	174.7	0.5347	0.0000	OK
30 minute winter	S93	26	89.799	0.325	196.9	0.0000	0.0000	OK
15 minute winter	S73	10	90.587	0.114	23.3	0.1638	0.0000	OK
15 minute winter	S74	11	90.450	0.198	60.6	0.2834	0.0000	OK
30 minute winter	S71	29	90.420	0.323	68.4	0.4618	0.0000	OK
30 minute winter	S72	29	90.414	0.471	522.9	0.0000	0.0000	OK
15 minute winter	S75	10	90.562	0.109	19.0	0.1563	0.0000	OK
15 minute winter	S76	11	90.532	0.221	54.9	0.3167	0.0000	OK
15 minute winter	S77	11	90.541	0.154	19.0	0.2197	0.0000	OK
15 minute winter	S68	11	90.380	0.201	71.4	0.2879	0.0000	OK
30 minute winter	S69	31	90.363	0.330	70.4	0.4717	0.0000	OK
30 minute winter	S70	30	90.361	0.479	532.3	0.0000	0.0000	OK
30 minute winter	S65	23	90.378	0.077	10.0	0.1102	0.0000	OK
30 minute winter	S66	24	90.216	0.108	20.0	0.1546	0.0000	OK
30 minute winter	S67	31	90.048	0.217	29.8	0.3112	0.0000	OK
30 minute winter	S62	36	89.952	0.132	7.7	0.1890	0.0000	OK
30 minute winter	S63	36	89.951	0.171	23.0	0.2447	0.0000	OK
30 minute winter	S64	37	89.953	0.083	7.7	0.1193	0.0000	OK

Link Event (Upstream Depth)	US Node	Link	DS Node	Outflow (l/s)	Velocity (m/s)	Flow/Cap	Link Vol (m³)	Discharge Vol (m³)
30 minute winter	S57	1.031	S53	481.8	0.512	0.062	51.7530	
30 minute winter	S53	1.032	S54 HW	480.5	1.485	0.518	7.6905	
240 minute winter	S54 HW	Flow through pond	S62 HW	174.7	0.065	0.013	1635.9218	
240 minute winter	S62 HW	20.000	Basin A Flow Control	37.7	0.272	0.022	2.9416	
30 minute winter	S93	16.014	S54 HW	192.7	1.451	0.701	2.6222	
15 minute winter	S73	7.000_1	S74	22.7	0.636	0.316	1.9390	
15 minute winter	S74	7.001_1	S71	59.2	0.899	0.526	3.1877	
30 minute winter	S71	7.002	S72	66.0	0.929	0.587	5.1053	
30 minute winter	S72	1.025_1	S60	512.7	0.461	0.182	54.5654	
15 minute winter	S75	11.000	S76	18.5	0.484	0.290	1.6803	
15 minute winter	S76	11.001	S68	53.6	1.011	0.839	2.0917	
15 minute winter	S77	12.000	S76	17.7	0.458	0.277	1.0469	
15 minute winter	S68	11.002	S69	71.0	0.997	0.383	3.5902	
30 minute winter	S69	11.003	S70	69.4	1.131	0.372	6.3669	
30 minute winter	S70	1.027_1	S50	529.8	0.601	0.188	109.0118	
30 minute winter	S65	13.000	S66	10.0	0.715	0.139	0.3916	
30 minute winter	S66	13.001	S67	19.9	0.797	0.177	1.8365	
30 minute winter	S67	13.002	S55	29.3	0.734	0.274	9.0245	
30 minute winter	S62	15.000	S63	7.6	0.311	0.160	0.7547	
30 minute winter	S63	14.001	S52	22.9	0.829	0.388	1.4529	
30 minute winter	S64	14.000	S63	7.7	0.342	0.163	1.3945	

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Results for 30 year Critical Storm Duration. Lowest mass balance: 99.78%

Node Event	US Node	Peak (mins)	Level (m)	Depth (m)	Inflow (l/s)	Node Vol (m³)	Flood (m³)	Status
15 minute winter	6	15	91.134	0.050	3.3	0.0721	0.0000	OK
15 minute winter	7	15	91.127	0.304	18.3	0.4349	0.0000	OK
15 minute winter	8	15	91.123	0.416	166.4	0.5951	0.0000	OK
15 minute winter	30	15	91.058	0.289	40.8	0.4132	0.0000	OK
15 minute winter	9	15	91.051	0.461	209.3	0.6592	0.0000	OK
15 minute winter	10	16	90.986	0.453	217.2	0.8011	0.0000	OK
30 minute winter	40	21	90.833	0.390	36.3	0.4411	0.0000	SURCHARGED
15 minute winter	11	16	90.845	0.566	247.5	0.9993	0.0000	OK
15 minute winter	50	16	90.788	0.248	47.0	0.2807	0.0000	OK
15 minute winter	S79	16	90.804	0.565	241.2	0.9977	0.0000	OK
15 minute winter	12	16	90.762	0.563	288.9	0.9952	0.0000	OK
15 minute winter	S78	16	90.706	0.536	288.2	0.9474	0.0000	OK
15 minute winter	13	16	90.570	0.523	312.3	0.9239	0.0000	OK
15 minute winter	14	16	90.456	0.462	322.5	0.8156	0.0000	OK
15 minute winter	15	16	90.263	0.457	350.7	1.1639	0.0000	OK
15 minute winter	60	10	90.681	0.351	91.6	0.3970	0.0000	SURCHARGED
30 minute winter	16	25	90.019	0.426	388.6	1.0840	0.0000	OK
30 minute winter	17	27	89.969	0.449	387.7	1.1429	0.0000	OK
	18							
240 minute winter	Basin A Flow Control	260	89.890	0.699	39.6	1.7790	0.0000	SURCHARGED
60 minute summer	Basin A Outfall	1441	89.290	0.400	38.7	0.0000	0.0000	OK
120 minute summer	Basin A to Swale	202	89.320	0.214	38.9	0.0000	0.0000	OK
120 minute summer	S87	202	89.317	0.237	38.9	0.0000	0.0000	OK
120 minute summer	S88	202	89.315	0.257	38.9	0.0000	0.0000	OK
240 minute winter	S89	1444	89.294	0.331	38.9	0.0000	0.0000	OK

Link Event (Upstream Depth)	US Node	Link	DS Node	Outflow (l/s)	Velocity (m/s)	Flow/Cap	Link Vol (m³)	Discharge Vol (m³)
15 minute winter	6	1.005	7	-3.3	0.273	-0.029	2.3102	
15 minute winter	7	1.006	8	28.1	0.352	0.179	6.4730	
15 minute winter	8	1.007	9	164.8	1.175	1.047	2.6013	
15 minute winter	30	2.000	9	40.6	0.872	0.363	0.8658	
15 minute winter	9	1.008	10	207.7	1.103	0.920	5.1649	
15 minute winter	10	1.009	11	211.8	1.144	0.938	16.6244	
30 minute winter	40	8.000_1	S74	37.1	0.684	0.668	3.2236	
15 minute winter	11	1.010	S79	241.2	0.912	0.850	6.3934	
15 minute winter	50	5.001	12	46.5	1.018	0.652	0.6313	
15 minute winter	S79	1.211	12	239.2	0.891	0.845	6.4137	
15 minute winter	12	1.011	S78	288.2	1.091	1.016	4.5478	
15 minute winter	S78	1.212	13	287.2	1.094	1.018	19.0100	
15 minute winter	13	1.012	14	311.5	1.259	1.097	7.5715	
15 minute winter	14	1.013	15	321.8	1.542	1.031	3.7976	
15 minute winter	15	1.014	16	350.4	1.425	0.765	11.1050	
15 minute winter	60	5.000	16	89.4	1.302	1.224	2.1793	
30 minute winter	16	1.015	17	387.7	1.475	0.292	4.9908	
30 minute winter	17	1.016	S93	359.9	0.560	0.072	66.9093	
240 minute winter	Basin A Flow Control	Hydro-Brake®	Basin A to Swale	38.9				
120 minute summer	Basin A to Swale	20.002	S87	38.9	0.145	0.014	10.0016	
120 minute summer	S87	20.003	S88	38.9	0.125	0.014	9.4973	
120 minute summer	S88	20.004	S89	38.9	0.436	0.632	3.9560	
240 minute winter	S89	20.005	S92	38.9	0.137	0.017	5.1504	

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Results for 30 year Critical Storm Duration. Lowest mass balance: 99.78%

Node Event	US Node	Peak (mins)	Level (m)	Depth (m)	Inflow (l/s)	Node Vol (m³)	Flood (m³)	Status
240 minute winter	S92	1444	89.294	0.341	38.9	0.0000	0.0000	OK
240 minute winter	S90	1444	89.294	0.350	38.9	0.0000	0.0000	OK
30 minute winter	S12	23	97.648	0.081	9.5	0.1160	0.0000	OK
30 minute winter	S13	23	97.509	0.089	19.0	0.1010	0.0000	OK
30 minute winter	S14	23	97.141	0.099	28.5	0.1417	0.0000	OK
30 minute winter	S15	22	96.645	0.126	61.4	0.1810	0.0000	OK
30 minute winter	S16	27	96.320	0.450	94.4	0.6433	0.0000	SURCHARGED
30 minute winter	S41	23	97.602	0.099	23.5	0.1413	0.0000	OK
30 minute winter	S42	27	97.551	0.391	47.0	0.5596	0.0000	SURCHARGED
30 minute winter	S43	27	97.527	0.517	72.0	0.7395	0.0000	SURCHARGED
30 minute winter	S37	23	98.240	0.086	23.5	0.1224	0.0000	OK
30 minute winter	S38	23	97.709	0.128	47.0	0.1831	0.0000	OK
30 minute winter	S39	27	97.506	0.156	70.5	0.2230	0.0000	OK
30 minute winter	S40	27	97.380	0.660	164.5	0.7469	0.0000	SURCHARGED
30 minute winter	S17	27	96.234	0.986	285.0	1.4105	0.0000	SURCHARGED
30 minute winter	S18	27	95.993	1.196	300.8	1.7122	0.0000	SURCHARGED
30 minute winter	S35	27	95.735	0.288	23.5	0.4122	0.0000	OK
30 minute winter	S36	27	95.713	0.871	46.9	1.2467	0.0000	SURCHARGED
30 minute winter	S19	27	95.705	1.158	338.3	1.6564	0.0000	SURCHARGED
30 minute winter	S34	27	95.090	0.548	13.0	0.7835	0.0000	SURCHARGED
30 minute winter	S20	27	95.064	0.985	366.3	1.4090	0.0000	SURCHARGED
30 minute winter	S21	27	94.708	1.245	388.2	1.7813	0.0000	SURCHARGED
30 minute winter	S22	27	94.400	1.085	410.2	1.5523	0.0000	SURCHARGED
30 minute winter	S23	26	93.792	0.847	432.4	1.2119	0.0000	SURCHARGED

Link Event (Upstream Depth)	US Node	Link	DS Node	Outflow (l/s)	Velocity (m/s)	Flow/Cap	Link Vol (m³)	Discharge Vol (m³)
240 minute winter	S92	20.006	S90	38.9	0.130	0.017	4.9608	
240 minute winter	S90	20.007	Basin A Outfall	38.9	0.585	0.636	3.0938	2996.0
30 minute winter	S12	1.000	S13	9.5	0.691	0.276	0.4574	
30 minute winter	S13	1.001	S14	19.0	1.005	0.194	0.9138	
30 minute winter	S14	1.002	S15	28.4	1.439	0.223	0.6757	
30 minute winter	S15	1.003	S16	61.4	1.562	0.251	3.0911	
30 minute winter	S16	1.004	S17	92.0	1.524	0.389	4.3286	
30 minute winter	S41	2.000_1	S42	23.5	0.871	0.238	1.9495	
30 minute winter	S42	2.001	S43	50.0	1.116	0.387	0.7840	
30 minute winter	S43	2.002	S40	70.9	1.225	0.753	2.8340	
30 minute winter	S37	3.000	S38	23.5	1.047	0.181	0.9458	
30 minute winter	S38	3.001	S39	47.0	1.504	0.324	0.4293	
30 minute winter	S39	3.002	S40	70.5	1.425	0.483	1.9711	
30 minute winter	S40	2.003	S17	160.4	2.608	0.965	4.1939	
30 minute winter	S17	1.005_1	S18	269.0	2.195	0.705	4.2865	
30 minute winter	S18	1.006_1	S19	291.5	1.840	0.921	4.1328	
30 minute winter	S35	4.000	S36	27.8	1.284	0.230	3.1291	
30 minute winter	S36	4.001	S19	50.5	1.016	0.178	1.2063	
30 minute winter	S19	1.007_1	S20	340.8	2.371	1.075	7.7135	
30 minute winter	S34	5.000_1	S20	16.5	0.811	0.439	1.7932	
30 minute winter	S20	1.008_1	S21	368.1	2.324	0.597	2.7004	
30 minute winter	S21	1.009_1	S22	387.9	2.449	1.032	1.7386	
30 minute winter	S22	1.010_1	S23	410.1	2.588	1.110	4.4942	
30 minute winter	S23	1.011_1	S24	432.1	2.727	1.142	2.0721	

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Results for 30 year Critical Storm Duration. Lowest mass balance: 99.78%

Node Event	US Node	Peak (mins)	Level (m)	Depth (m)	Inflow (l/s)	Node Vol (m³)	Flood (m³)	Status
30 minute winter	S24	26	93.385	0.693	454.3	1.2247	0.0000	SURCHARGED
30 minute winter	S25	26	92.696	0.976	467.8	1.7247	0.0000	SURCHARGED
30 minute winter	S26	25	91.852	1.032	483.0	2.6255	0.0000	SURCHARGED
30 minute winter	S27	25	91.797	1.134	501.8	2.8868	0.0000	SURCHARGED
30 minute winter	S28	25	91.684	1.151	521.0	2.9285	0.0000	SURCHARGED
30 minute winter	S29	25	91.568	1.101	557.8	2.8008	0.0000	SURCHARGED
30 minute winter	S30	25	91.426	1.029	594.6	2.6189	0.0000	SURCHARGED
30 minute winter	S31	25	91.244	1.004	631.6	2.5562	0.0000	SURCHARGED
30 minute winter	S32	25	91.136	0.968	668.7	2.4640	0.0000	SURCHARGED
30 minute winter	S44	23	94.785	0.184	183.6	0.2626	0.0000	OK
30 minute winter	S45	22	92.943	0.213	229.1	0.2404	0.0000	OK
30 minute winter	S46	23	91.476	0.246	275.8	0.2780	0.0000	OK
30 minute winter	S33	24	91.071	0.921	924.0	2.3450	0.0000	SURCHARGED
30 minute winter	S47	25	90.953	0.844	945.5	1.2076	0.0000	OK
30 minute winter	S48	25	90.788	0.731	951.0	0.8263	0.0000	OK
30 minute winter	S49	26	90.579	0.588	950.6	0.8421	0.0000	OK
240 minute winter	S58	240	90.975	0.455	12.1	0.6514	0.0000	OK
240 minute winter	S61	244	90.975	0.499	12.1	0.7140	0.0000	OK
240 minute winter	S59 Flow Control	244	90.975	0.600	21.8	0.8580	0.0000	SURCHARGED
30 minute winter	S60	27	90.519	0.624	998.3	0.8923	0.0000	OK
30 minute winter	S50	30	90.228	0.466	1040.4	0.6673	0.0000	OK
30 minute winter	S55	32	90.191	0.463	1062.5	0.6620	0.0000	OK
30 minute winter	S51	34	90.159	0.479	1052.2	0.6850	0.0000	OK
30 minute winter	S52	35	90.135	0.562	1034.3	0.8036	0.0000	OK
30 minute winter	S56	35	90.124	0.571	954.9	0.8165	0.0000	OK

Link Event (Upstream Depth)	US Node	Link	DS Node	Outflow (l/s)	Velocity (m/s)	Flow/Cap	Link Vol (m³)	Discharge Vol (m³)
30 minute winter	S24	1.012_1	S25	453.6	2.420	0.792	15.0527	
30 minute winter	S25	1.013_1	S26	465.5	2.322	1.045	17.2472	
30 minute winter	S26	1.014_1	S27	484.3	1.385	0.399	7.1572	
30 minute winter	S27	1.015_1	S28	503.1	1.143	0.809	22.4035	
30 minute winter	S28	1.016_1	S29	521.8	1.186	1.140	20.8705	
30 minute winter	S29	1.017	S30	558.6	1.269	1.230	22.4677	
30 minute winter	S30	1.018_1	S31	595.6	1.354	1.305	26.0833	
30 minute winter	S31	1.019	S32	632.7	1.187	1.015	24.5705	
30 minute winter	S32	1.020	S33	669.5	1.256	1.214	7.8243	
30 minute winter	S44	6.000	S45	183.4	2.761	0.352	4.8619	
30 minute winter	S45	6.001	S46	229.3	2.847	0.445	4.7907	
30 minute winter	S46	6.002	S33	275.1	3.305	0.456	2.2455	
30 minute winter	S33	1.021	S47	923.7	1.465	1.346	21.3265	
30 minute winter	S47	1.022	S48	940.7	1.605	1.349	24.4459	
30 minute winter	S48	1.023	S49	950.6	1.927	1.119	17.7707	
30 minute winter	S49	1.024	S72	945.5	0.556	0.140	82.8995	
240 minute winter	S58	7.000	S59 Flow Control	6.5	0.087	0.001	104.7053	
240 minute winter	S61	8.000	S59 Flow Control	6.4	0.116	0.001	73.3530	
240 minute winter	S59 Flow Control	Hydro-Brake®	S60	2.6				
30 minute winter	S60	1.025	S70	993.7	0.549	0.144	22.6390	
30 minute winter	S50	1.026	S55	1019.0	0.949	0.150	37.4871	
30 minute winter	S55	1.027	S51	1052.2	1.024	0.155	54.3031	
30 minute winter	S51	1.028	S52	1006.7	0.958	0.149	143.3748	
30 minute winter	S52	1.029	S56	954.9	0.898	0.139	29.8475	
30 minute winter	S56	1.030	S57	915.2	0.871	0.135	79.0100	

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Results for 30 year Critical Storm Duration. Lowest mass balance: 99.78%

Node Event	US Node	Peak (mins)	Level (m)	Depth (m)	Inflow (l/s)	Node Vol (m³)	Flood (m³)	Status
30 minute winter	S57	35	90.112	0.607	915.2	0.8692	0.0000	OK
30 minute winter	S53	35	90.105	0.665	895.4	0.9514	0.0000	OK
240 minute winter	S54 HW	260	89.890	0.662	593.6	1.8771	0.0000	OK
240 minute winter	S62 HW	260	89.890	0.665	316.7	0.9518	0.0000	OK
30 minute winter	S93	27	89.962	0.488	359.9	0.0000	0.0000	OK
30 minute winter	S73	21	90.784	0.311	34.0	0.4446	0.0000	SURCHARGED
30 minute winter	S74	21	90.765	0.513	96.4	0.7345	0.0000	SURCHARGED
30 minute winter	S71	21	90.656	0.559	119.5	0.7995	0.0000	SURCHARGED
30 minute winter	S72	27	90.573	0.630	1023.8	0.0000	0.0000	OK
15 minute winter	S75	11	90.932	0.479	36.0	0.6857	0.0000	SURCHARGED
15 minute winter	S76	11	90.883	0.572	101.9	0.8186	0.0000	SURCHARGED
15 minute winter	S77	11	90.911	0.524	36.0	0.7499	0.0000	SURCHARGED
30 minute winter	S68	28	90.515	0.336	104.5	0.4813	0.0000	OK
30 minute winter	S69	27	90.515	0.482	130.5	0.6898	0.0000	SURCHARGED
30 minute winter	S70	27	90.514	0.632	1046.1	0.0000	0.0000	OK
30 minute winter	S65	23	90.410	0.109	19.1	0.1556	0.0000	OK
30 minute winter	S66	21	90.259	0.151	38.2	0.2155	0.0000	OK
30 minute winter	S67	29	90.196	0.365	57.7	0.5221	0.0000	OK
30 minute winter	S62	34	90.136	0.316	14.7	0.4519	0.0000	SURCHARGED
30 minute winter	S63	34	90.136	0.356	43.7	0.5094	0.0000	SURCHARGED
30 minute winter	S64	34	90.136	0.266	14.7	0.3806	0.0000	OK

Link Event (Upstream Depth)	US Node	Link	DS Node	Outflow (l/s)	Velocity (m/s)	Flow/Cap	Link Vol (m³)	Discharge Vol (m³)
30 minute winter	S57	1.031	S53	895.4	0.569	0.115	92.3948	
30 minute winter	S53	1.032	S54 HW	892.3	1.810	0.963	11.6895	
240 minute winter	S54 HW	Flow through pond	S62 HW	316.7	0.083	0.024	3002.5708	
240 minute winter	S62 HW	20.000	Basin A Flow Control	39.6	0.271	0.023	5.9730	
30 minute winter	S93	16.014	S54 HW	347.9	1.786	1.266	3.8078	
30 minute winter	S73	7.000_1	S74	33.3	0.658	0.464	3.7005	
30 minute winter	S74	7.001_1	S71	87.4	0.924	0.776	5.3430	
30 minute winter	S71	7.002	S72	112.3	1.062	1.000	5.3320	
30 minute winter	S72	1.025_1	S60	995.7	0.560	0.354	87.0343	
15 minute winter	S75	11.000	S76	35.0	0.528	0.549	3.0102	
15 minute winter	S76	11.001	S68	102.1	1.450	1.598	2.7807	
15 minute winter	S77	12.000	S76	33.1	0.483	0.518	1.6028	
30 minute winter	S68	11.002	S69	102.9	1.010	0.555	6.2697	
30 minute winter	S69	11.003	S70	116.0	1.326	0.622	7.1323	
30 minute winter	S70	1.027_1	S50	1040.4	0.731	0.370	176.2606	
30 minute winter	S65	13.000	S66	19.1	0.853	0.266	0.6274	
30 minute winter	S66	13.001	S67	38.6	0.916	0.344	4.0057	
30 minute winter	S67	13.002	S55	50.0	0.783	0.467	13.5830	
30 minute winter	S62	15.000	S63	14.5	0.337	0.303	1.4898	
30 minute winter	S63	14.001	S52	42.6	0.984	0.723	1.9754	
30 minute winter	S64	14.000	S63	14.5	0.389	0.307	3.3405	

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Results for 100 year Critical Storm Duration. Lowest mass balance: 99.78%

Node Event	US Node	Peak (mins)	Level (m)	Depth (m)	Inflow (l/s)	Node Vol (m³)	Flood (m³)	Status
15 minute winter	6	15	91.787	0.703	43.6	1.0058	0.0000	SURCHARGED
15 minute winter	7	15	91.811	0.988	46.4	1.4137	0.0000	SURCHARGED
15 minute winter	8	15	91.826	1.119	210.5	1.6018	0.0000	SURCHARGED
15 minute winter	30	15	91.731	0.962	53.0	1.3763	0.0000	SURCHARGED
15 minute winter	9	15	91.715	1.125	267.9	1.6105	0.0000	SURCHARGED
15 minute winter	10	15	91.597	1.064	279.2	1.8804	0.0000	SURCHARGED
15 minute winter	40	11	91.239	0.796	45.6	0.9002	0.0000	SURCHARGED
15 minute winter	11	15	91.272	0.993	322.3	1.7542	0.0000	SURCHARGED
15 minute winter	50	15	91.147	0.607	61.0	0.6861	0.0000	SURCHARGED
15 minute winter	S79	15	91.185	0.946	320.5	1.6720	0.0000	SURCHARGED
15 minute winter	12	15	91.099	0.900	383.1	1.5909	0.0000	SURCHARGED
15 minute winter	S78	15	91.000	0.830	381.8	1.4659	0.0000	SURCHARGED
15 minute winter	13	15	90.730	0.683	417.1	1.2062	0.0000	SURCHARGED
15 minute winter	14	15	90.563	0.569	433.0	1.0060	0.0000	OK
15 minute winter	15	16	90.354	0.548	475.7	1.3952	0.0000	OK
15 minute winter	60	10	90.874	0.544	119.0	0.6156	0.0000	SURCHARGED
30 minute winter	16	25	90.123	0.530	526.6	1.3480	0.0000	OK
30 minute winter	17	26	90.083	0.563	524.8	1.4329	0.0000	OK
	18							
240 minute winter	Basin A Flow Control	260	90.056	0.865	40.1	2.2020	0.0000	SURCHARGED
30 minute summer	Basin A Outfall	1441	89.290	0.400	38.8	0.0000	0.0000	OK
240 minute winter	Basin A to Swale	1452	89.358	0.252	38.9	0.0000	0.0000	OK
240 minute winter	S87	1452	89.357	0.277	38.9	0.0000	0.0000	OK
240 minute winter	S88	1452	89.356	0.298	38.8	0.0000	0.0000	OK
240 minute winter	S89	1456	89.310	0.347	38.8	0.0000	0.0000	OK

Link Event (Upstream Depth)	US Node	Link	DS Node	Outflow (l/s)	Velocity (m/s)	Flow/Cap	Link Vol (m³)	Discharge Vol (m³)
15 minute winter	6	1.005	7	-43.6	-0.406	-0.388	6.4310	
15 minute winter	7	1.006	8	72.8	0.498	0.463	7.6836	
15 minute winter	8	1.007	9	208.7	1.318	1.326	2.7693	
15 minute winter	30	2.000	9	56.6	0.898	0.505	1.0121	
15 minute winter	9	1.008	10	267.5	1.239	1.185	5.6019	
15 minute winter	10	1.009	11	278.5	1.289	1.233	17.5952	
15 minute winter	40	8.000_1	S74	47.2	0.670	0.850	3.2236	
15 minute winter	11	1.010	S79	320.5	1.138	1.130	6.5436	
15 minute winter	50	5.001	12	61.1	1.073	0.857	0.6955	
15 minute winter	S79	1.211	12	318.6	1.131	1.126	6.5727	
15 minute winter	12	1.011	S78	381.8	1.356	1.346	4.7409	
15 minute winter	S78	1.212	13	383.1	1.360	1.358	20.3470	
15 minute winter	13	1.012	14	416.8	1.480	1.468	8.5659	
15 minute winter	14	1.013	15	434.3	1.729	1.392	4.5117	
15 minute winter	15	1.014	16	476.5	1.575	1.040	13.6085	
15 minute winter	60	5.000	16	116.9	1.664	1.601	2.2898	
30 minute winter	16	1.015	17	524.8	1.559	0.395	6.5714	
30 minute winter	17	1.016	S93	467.0	0.574	0.093	98.8638	
240 minute winter	Basin A Flow Control	Hydro-Brake®	Basin A to Swale	38.9				
240 minute winter	Basin A to Swale	20.002	S87	38.9	0.129	0.014	12.5695	
240 minute winter	S87	20.003	S88	38.8	0.101	0.014	11.8193	
240 minute winter	S88	20.004	S89	38.8	0.441	0.630	4.6513	
240 minute winter	S89	20.005	S92	38.7	0.137	0.017	5.5658	

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Results for 100 year Critical Storm Duration. Lowest mass balance: 99.78%

Node Event	US Node	Peak (mins)	Level (m)	Depth (m)	Inflow (l/s)	Node Vol (m³)	Flood (m³)	Status
240 minute winter	S92	1456	89.310	0.357	38.7	0.0000	0.0000	OK
240 minute winter	S90	1456	89.310	0.366	38.6	0.0000	0.0000	OK
30 minute winter	S12	23	97.662	0.095	12.5	0.1356	0.0000	OK
30 minute winter	S13	22	97.523	0.103	25.0	0.1169	0.0000	OK
30 minute winter	S14	25	97.310	0.268	37.7	0.3830	0.0000	OK
30 minute winter	S15	25	97.269	0.750	80.5	1.0733	0.0000	SURCHARGED
30 minute winter	S16	25	97.191	1.321	117.7	1.8909	0.0000	SURCHARGED
30 minute winter	S41	24	98.772	1.269	30.8	1.8165	0.0000	SURCHARGED
30 minute winter	S42	24	98.729	1.569	61.9	2.2448	0.0000	FLOOD RISK
30 minute winter	S43	24	98.671	1.661	92.2	2.3769	0.3797	FLOOD
30 minute winter	S37	24	98.783	0.629	30.8	0.9004	0.0000	SURCHARGED
30 minute winter	S38	24	98.743	1.162	64.9	1.6622	0.0000	SURCHARGED
30 minute winter	S39	24	98.677	1.327	93.0	1.8985	0.0000	FLOOD RISK
30 minute winter	S40	24	98.372	1.652	207.4	1.8684	13.0215	FLOOD
30 minute winter	S17	24	97.024	1.776	315.3	2.5415	1.2241	FLOOD
30 minute winter	S18	23	96.680	1.883	349.8	2.6946	12.7575	FLOOD
30 minute winter	S35	22	96.432	0.985	30.8	1.4094	0.0000	SURCHARGED
30 minute winter	S36	22	96.356	1.514	63.9	2.1660	0.0000	FLOOD RISK
30 minute winter	S19	22	96.334	1.787	385.9	2.5572	23.8247	FLOOD
30 minute winter	S34	22	95.798	1.256	17.1	1.7975	0.0000	FLOOD RISK
30 minute winter	S20	22	95.729	1.650	387.4	2.3612	4.9318	FLOOD
30 minute winter	S21	23	95.404	1.941	407.4	2.7782	0.0000	FLOOD RISK
30 minute winter	S22	23	95.107	1.792	427.3	2.5642	0.0000	FLOOD RISK
30 minute winter	S23	23	94.480	1.535	453.5	2.1971	0.0000	SURCHARGED

Link Event (Upstream Depth)	US Node	Link	DS Node	Outflow (l/s)	Velocity (m/s)	Flow/Cap	Link Vol (m³)	Discharge Vol (m³)
240 minute winter	S92	20.006	S90	38.6	0.130	0.017	5.3454	
240 minute winter	S90	20.007	Basin A Outfall	38.6	0.584	0.631	3.1560	3519.1
30 minute winter	S12	1.000	S13	12.5	0.746	0.363	0.5582	
30 minute winter	S13	1.001	S14	25.2	1.099	0.257	2.1052	
30 minute winter	S14	1.002	S15	38.5	1.530	0.302	2.3387	
30 minute winter	S15	1.003	S16	77.1	1.591	0.316	4.8109	
30 minute winter	S16	1.004	S17	117.8	1.525	0.498	4.3286	
30 minute winter	S41	2.000_1	S42	31.5	0.892	0.320	3.0642	
30 minute winter	S42	2.001	S43	61.8	1.128	0.478	0.7840	
30 minute winter	S43	2.002	S40	87.4	1.464	0.928	2.8340	
30 minute winter	S37	3.000	S38	34.5	1.087	0.265	2.9487	
30 minute winter	S38	3.001	S39	62.6	1.527	0.432	0.9602	
30 minute winter	S39	3.002	S40	90.4	1.458	0.620	2.5853	
30 minute winter	S40	2.003	S17	159.7	2.602	0.961	4.1939	
30 minute winter	S17	1.005_1	S18	307.2	2.176	0.805	4.2865	
30 minute winter	S18	1.006_1	S19	312.1	1.970	0.986	4.1328	
30 minute winter	S35	4.000	S36	33.1	1.296	0.274	3.1498	
30 minute winter	S36	4.001	S19	62.0	1.038	0.218	1.2063	
30 minute winter	S19	1.007_1	S20	358.6	2.347	1.132	7.7135	
30 minute winter	S34	5.000_1	S20	17.1	0.815	0.455	1.7932	
30 minute winter	S20	1.008_1	S21	391.9	2.474	0.636	2.7004	
30 minute winter	S21	1.009_1	S22	411.8	2.599	1.096	1.7386	
30 minute winter	S22	1.010_1	S23	431.5	2.724	1.168	4.4942	
30 minute winter	S23	1.011_1	S24	451.0	2.847	1.192	2.0721	

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Results for 100 year Critical Storm Duration. Lowest mass balance: 99.78%

Node Event	US Node	Peak (mins)	Level (m)	Depth (m)	Inflow (l/s)	Node Vol (m³)	Flood (m³)	Status
30 minute winter	S24	23	94.044	1.352	472.2	2.3889	0.0000	SURCHARGED
30 minute winter	S25	23	93.282	1.562	496.8	2.7596	0.0000	SURCHARGED
30 minute winter	S26	23	92.314	1.494	521.9	3.8028	0.0000	SURCHARGED
30 minute winter	S27	23	92.250	1.587	548.5	4.0396	0.0000	SURCHARGED
30 minute winter	S28	23	92.113	1.580	575.1	4.0214	0.0000	SURCHARGED
30 minute winter	S29	23	91.970	1.503	625.7	3.8242	0.0000	SURCHARGED
30 minute winter	S30	23	91.791	1.394	676.7	3.5466	0.0000	SURCHARGED
30 minute winter	S31	23	91.557	1.317	727.7	3.3529	0.0000	SURCHARGED
30 minute winter	S32	23	91.416	1.248	778.8	3.1773	0.0000	SURCHARGED
30 minute winter	S44	23	94.817	0.216	240.3	0.3092	0.0000	OK
30 minute winter	S45	21	92.974	0.244	299.8	0.2762	0.0000	OK
30 minute winter	S46	23	91.611	0.381	362.0	0.4310	0.0000	OK
30 minute winter	S33	23	91.328	1.178	1137.8	2.9988	0.0000	SURCHARGED
30 minute winter	S47	23	91.139	1.030	1173.6	1.4741	0.0000	SURCHARGED
30 minute winter	S48	24	90.915	0.858	1187.2	0.9709	0.0000	OK
30 minute winter	S49	25	90.651	0.660	1185.1	0.9441	0.0000	OK
240 minute winter	S58	240	91.085	0.565	17.3	0.8092	0.0000	OK
240 minute winter	S61	244	91.085	0.609	15.8	0.8711	0.0000	OK
240 minute winter	S59 Flow Control	244	91.085	0.710	28.5	1.0161	0.0000	SURCHARGED
30 minute winter	S60	25	90.589	0.694	1270.4	0.9929	0.0000	OK
30 minute winter	S50	31	90.293	0.531	1347.0	0.7603	0.0000	OK
60 minute winter	S55	48	90.267	0.539	1268.3	0.7709	0.0000	OK
60 minute winter	S51	49	90.243	0.563	1244.7	0.8053	0.0000	OK
30 minute winter	S52	34	90.223	0.650	1338.0	0.9308	0.0000	OK
30 minute winter	S56	34	90.214	0.661	1205.2	0.9454	0.0000	OK

Link Event (Upstream Depth)	US Node	Link	DS Node	Outflow (l/s)	Velocity (m/s)	Flow/Cap	Link Vol (m³)	Discharge Vol (m³)
30 minute winter	S24	1.012_1	S25	473.0	2.438	0.825	15.0527	
30 minute winter	S25	1.013_1	S26	499.0	2.342	1.120	17.2472	
30 minute winter	S26	1.014_1	S27	524.2	1.395	0.432	7.1572	
30 minute winter	S27	1.015_1	S28	550.1	1.250	0.884	22.4035	
30 minute winter	S28	1.016_1	S29	576.5	1.310	1.260	20.8705	
30 minute winter	S29	1.017	S30	627.0	1.425	1.380	22.4677	
30 minute winter	S30	1.018_1	S31	677.5	1.540	1.485	26.0833	
30 minute winter	S31	1.019	S32	728.5	1.367	1.169	24.5705	
30 minute winter	S32	1.020	S33	779.5	1.462	1.414	7.8243	
30 minute winter	S44	6.000	S45	240.2	2.987	0.461	5.8757	
30 minute winter	S45	6.001	S46	300.9	2.933	0.584	6.8090	
30 minute winter	S46	6.002	S33	357.5	3.330	0.593	2.7434	
30 minute winter	S33	1.021	S47	1137.6	1.795	1.658	21.6028	
30 minute winter	S47	1.022	S48	1173.5	1.852	1.682	26.2919	
30 minute winter	S48	1.023	S49	1185.1	2.116	1.395	20.1357	
30 minute winter	S49	1.024	S72	1168.8	0.589	0.172	99.6251	
240 minute winter	S58	7.000	S59 Flow Control	8.7	0.095	0.001	142.3332	
240 minute winter	S61	8.000	S59 Flow Control	11.7	0.122	0.001	98.9305	
240 minute winter	S59 Flow Control	Hydro-Brake®	S60	2.6				
30 minute winter	S60	1.025	S70	1262.6	0.586	0.183	27.0002	
30 minute winter	S50	1.026	S55	1313.0	1.007	0.193	47.0876	
60 minute winter	S55	1.027	S51	1244.7	1.019	0.183	69.9729	
60 minute winter	S51	1.028	S52	1175.9	0.888	0.174	184.1200	
30 minute winter	S52	1.029	S56	1205.2	0.950	0.176	38.0069	
30 minute winter	S56	1.030	S57	1142.2	0.908	0.169	100.1447	

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Results for 100 year Critical Storm Duration. Lowest mass balance: 99.78%

Node Event	US Node	Peak (mins)	Level (m)	Depth (m)	Inflow (l/s)	Node Vol (m³)	Flood (m³)	Status
30 minute winter	S57	34	90.204	0.699	1142.2	1.0000	0.0000	OK
30 minute winter	S53	34	90.197	0.757	1115.1	1.0830	0.0000	OK
240 minute winter	S54 HW	260	90.056	0.828	772.6	2.3481	0.0000	OK
240 minute winter	S62 HW	260	90.056	0.831	406.8	1.1896	0.0000	OK
30 minute winter	S93	26	90.081	0.607	467.0	0.0000	0.0000	SURCHARGED
15 minute winter	S73	11	91.298	0.825	57.5	1.1803	0.0000	SURCHARGED
15 minute winter	S74	11	91.173	0.921	143.6	1.3178	0.0000	SURCHARGED
30 minute winter	S71	20	90.891	0.794	169.0	1.1369	0.0000	SURCHARGED
30 minute winter	S72	25	90.644	0.701	1290.7	0.0000	0.0000	OK
15 minute winter	S75	11	91.338	0.885	46.7	1.2666	0.0000	SURCHARGED
15 minute winter	S76	11	91.259	0.948	132.3	1.3571	0.0000	SURCHARGED
15 minute winter	S77	11	91.307	0.920	46.7	1.3168	0.0000	SURCHARGED
30 minute summer	S68	19	90.675	0.496	155.2	0.7104	0.0000	SURCHARGED
30 minute winter	S69	22	90.607	0.574	166.8	0.8217	0.0000	SURCHARGED
30 minute winter	S70	25	90.583	0.701	1354.1	0.0000	0.0000	OK
30 minute winter	S65	23	90.428	0.127	25.0	0.1810	0.0000	OK
30 minute winter	S66	27	90.310	0.202	50.0	0.2896	0.0000	OK
30 minute winter	S67	28	90.288	0.457	74.9	0.6536	0.0000	SURCHARGED
60 minute winter	S62	47	90.244	0.424	16.2	0.6065	0.0000	SURCHARGED
60 minute winter	S63	47	90.242	0.462	46.3	0.6612	0.0000	SURCHARGED
60 minute winter	S64	46	90.246	0.376	16.2	0.5384	0.0000	SURCHARGED

Link Event (Upstream Depth)	US Node	Link	DS Node	Outflow (l/s)	Velocity (m/s)	Flow/Cap	Link Vol (m³)	Discharge Vol (m³)
30 minute winter	S57	1.031	S53	1115.1	0.592	0.144	115.6779	
30 minute winter	S53	1.032	S54 HW	1111.6	1.956	1.199	13.4460	
240 minute winter	S54 HW	Flow through pond	S62 HW	406.8	0.093	0.031	3815.9817	
240 minute winter	S62 HW	20.000	Basin A Flow Control	40.1	0.272	0.023	7.5190	
30 minute winter	S93	16.014	S54 HW	435.7	2.033	1.585	4.0509	
15 minute winter	S73	7.000_1	S74	51.6	0.770	0.719	3.7005	
15 minute winter	S74	7.001_1	S71	141.1	1.279	1.253	5.3430	
30 minute winter	S71	7.002	S72	168.4	1.527	1.499	5.3320	
30 minute winter	S72	1.025_1	S60	1267.8	0.603	0.451	103.9491	
15 minute winter	S75	11.000	S76	44.2	0.628	0.695	3.0102	
15 minute winter	S76	11.001	S68	132.3	1.879	2.072	2.7807	
15 minute winter	S77	12.000	S76	44.1	0.626	0.691	1.6028	
30 minute summer	S68	11.002	S69	148.5	1.060	0.800	6.9582	
30 minute winter	S69	11.003	S70	167.0	1.412	0.896	7.1323	
30 minute winter	S70	1.027_1	S50	1347.0	0.791	0.479	211.9353	
30 minute winter	S65	13.000	S66	25.0	0.915	0.348	0.7658	
30 minute winter	S66	13.001	S67	49.9	0.932	0.444	5.4106	
30 minute winter	S67	13.002	S55	67.6	0.813	0.631	14.5351	
60 minute winter	S62	15.000	S63	15.3	0.327	0.319	1.4898	
60 minute winter	S63	14.001	S52	45.5	0.880	0.771	1.9754	
60 minute winter	S64	14.000	S63	15.1	0.382	0.321	3.4477	

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Results for 100 year +40% CC Critical Storm Duration. Lowest mass balance: 99.78%

Node Event	US Node	Peak (mins)	Level (m)	Depth (m)	Inflow (l/s)	Node Vol (m³)	Flood (m³)	Status
30 minute winter	6	19	92.424	1.340	57.5	1.9175	14.9574	FLOOD
15 minute winter	7	14	92.482	1.659	82.5	2.3738	0.0000	SURCHARGED
15 minute winter	8	12	92.472	1.765	309.4	2.5257	8.5502	FLOOD
30 minute winter	30	21	92.369	1.600	77.4	2.2898	0.0000	FLOOD RISK
30 minute winter	9	21	92.343	1.753	320.5	2.5086	0.0000	FLOOD RISK
30 minute winter	10	21	92.191	1.658	334.6	2.9292	0.0000	SURCHARGED
30 minute winter	40	20	91.905	1.462	66.6	1.6539	0.0000	SURCHARGED
30 minute winter	11	21	91.779	1.500	392.7	2.6502	0.0000	SURCHARGED
30 minute winter	50	21	91.647	1.107	89.1	1.2516	0.0000	SURCHARGED
30 minute winter	S79	21	91.660	1.421	386.1	2.5117	0.0000	SURCHARGED
30 minute winter	12	21	91.542	1.343	477.5	2.3726	0.0000	SURCHARGED
30 minute winter	S78	21	91.388	1.218	477.4	2.1522	0.0000	SURCHARGED
15 minute winter	13	13	90.971	0.924	530.0	1.6320	0.0000	SURCHARGED
15 minute winter	14	13	90.700	0.706	554.0	1.2483	0.0000	SURCHARGED
30 minute winter	15	23	90.468	0.662	614.5	1.6848	0.0000	OK
15 minute winter	60	10	91.354	1.024	166.5	1.1583	0.0000	SURCHARGED
240 minute winter	16	252	90.300	0.707	246.1	1.8004	0.0000	OK
240 minute winter	17	252	90.300	0.780	246.0	1.9863	0.0000	OK
	18							
240 minute winter	Basin A Flow Control	252	90.300	1.109	40.5	2.8236	0.0000	SURCHARGED
15 minute summer	Basin A Outfall	1441	89.290	0.400	38.8	0.0000	0.0000	OK
240 minute summer	Basin A to Swale	1652	89.363	0.257	38.9	0.0000	0.0000	OK
240 minute summer	S87	1652	89.361	0.281	38.9	0.0000	0.0000	OK
240 minute summer	S88	1652	89.361	0.303	38.9	0.0000	0.0000	OK
240 minute summer	S89	1656	89.312	0.349	38.9	0.0000	0.0000	OK

Link Event (Upstream Depth)	US Node	Link	DS Node	Outflow (l/s)	Velocity (m/s)	Flow/Cap	Link Vol (m³)	Discharge Vol (m³)
30 minute winter	6	1.005	7	-57.5	-0.596	-0.511	6.4310	
15 minute winter	7	1.006	8	76.2	0.484	0.485	7.6836	
15 minute winter	8	1.007	9	282.7	1.785	1.796	2.7693	
30 minute winter	30	2.000	9	78.7	0.814	0.703	1.0121	
30 minute winter	9	1.008	10	317.9	1.472	1.408	5.6019	
30 minute winter	10	1.009	11	330.1	1.528	1.462	17.5952	
30 minute winter	40	8.000_1	S74	69.7	0.990	1.256	3.2236	
30 minute winter	11	1.010	S79	386.1	1.371	1.362	6.5436	
30 minute winter	50	5.001	12	89.4	1.270	1.254	0.6955	
30 minute winter	S79	1.211	12	384.6	1.366	1.359	6.5727	
30 minute winter	12	1.011	S78	477.4	1.695	1.683	4.7409	
30 minute winter	S78	1.212	13	477.3	1.695	1.693	20.3470	
15 minute winter	13	1.012	14	530.0	1.882	1.866	8.6475	
15 minute winter	14	1.013	15	554.0	1.984	1.776	4.8858	
30 minute winter	15	1.014	16	614.7	1.711	1.341	16.8448	
15 minute winter	60	5.000	16	164.7	2.339	2.255	2.3537	
240 minute winter	16	1.015	17	246.0	1.161	0.185	9.1695	
240 minute winter	17	1.016	S93	243.5	0.448	0.048	175.1445	
240 minute winter	Basin A Flow Control	Hydro-Brake®	Basin A to Swale	38.9				
240 minute summer	Basin A to Swale	20.002	S87	38.9	0.131	0.014	12.8665	
240 minute summer	S87	20.003	S88	38.9	0.118	0.014	12.0790	
240 minute summer	S88	20.004	S89	38.9	0.433	0.632	4.6974	
240 minute summer	S89	20.005	S92	38.9	0.135	0.017	5.6077	

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Results for 100 year +40% CC Critical Storm Duration. Lowest mass balance: 99.78%

Node Event	US Node	Peak (mins)	Level (m)	Depth (m)	Inflow (l/s)	Node Vol (m³)	Flood (m³)	Status
240 minute summer	S92	1652	89.311	0.358	38.9	0.0000	0.0000	OK
240 minute summer	S90	1652	89.311	0.367	38.9	0.0000	0.0000	OK
30 minute winter	S12	25	97.689	0.122	17.5	0.1745	0.0000	OK
30 minute winter	S13	25	97.657	0.237	35.0	0.2686	0.0000	OK
30 minute winter	S14	24	97.618	0.576	68.7	0.8241	0.0000	SURCHARGED
30 minute winter	S15	24	97.532	1.013	109.7	1.4494	0.0000	SURCHARGED
30 minute winter	S16	24	97.372	1.502	169.4	2.1494	0.0000	FLOOD RISK
30 minute summer	S41	20	98.858	1.355	45.0	1.9396	0.0000	SURCHARGED
30 minute winter	S42	23	98.779	1.619	86.2	2.3169	0.0000	FLOOD RISK
30 minute winter	S43	19	98.671	1.661	129.3	2.3769	21.0755	FLOOD
30 minute winter	S37	20	99.072	0.918	43.1	1.3137	0.0000	SURCHARGED
30 minute winter	S38	23	98.992	1.411	86.2	2.0188	0.0000	FLOOD RISK
30 minute winter	S39	20	98.868	1.518	129.3	2.1723	3.6989	FLOOD
30 minute winter	S40	19	98.372	1.652	247.6	1.8684	56.7073	FLOOD
30 minute winter	S17	19	97.024	1.776	385.1	2.5415	42.0519	FLOOD
30 minute winter	S18	19	96.680	1.883	367.7	2.6946	36.7464	FLOOD
15 minute winter	S35	12	96.458	1.011	41.3	1.4463	0.0000	SURCHARGED
15 minute winter	S36	12	96.377	1.535	82.5	2.1968	0.0000	FLOOD RISK
30 minute winter	S19	19	96.334	1.787	415.8	2.5572	60.7757	FLOOD
30 minute winter	S34	19	95.848	1.306	24.3	1.8686	0.0000	FLOOD RISK
30 minute winter	S20	19	95.729	1.650	391.2	2.3612	56.4957	FLOOD
30 minute winter	S21	21	95.502	2.039	407.5	2.9178	3.3481	FLOOD
30 minute winter	S22	23	95.290	1.975	435.8	2.8266	0.0000	FLOOD RISK
30 minute winter	S23	23	94.800	1.855	464.8	2.6550	0.0000	FLOOD RISK

Link Event (Upstream Depth)	US Node	Link	DS Node	Outflow (l/s)	Velocity (m/s)	Flow/Cap	Link Vol (m³)	Discharge Vol (m³)
240 minute summer	S92	20.006	S90	38.9	0.128	0.017	5.3843	
240 minute summer	S90	20.007	Basin A Outfall	38.9	0.581	0.636	3.1619	3456.9
30 minute winter	S12	1.000	S13	17.5	0.804	0.509	1.0262	
30 minute winter	S13	1.001	S14	36.4	1.123	0.371	3.1476	
30 minute winter	S14	1.002	S15	53.5	1.546	0.420	2.4082	
30 minute winter	S15	1.003	S16	109.7	1.624	0.449	4.8109	
30 minute winter	S16	1.004	S17	169.4	1.536	0.716	4.3286	
30 minute summer	S41	2.000_1	S42	40.7	0.909	0.413	3.0642	
30 minute winter	S42	2.001	S43	86.2	1.224	0.667	0.7840	
30 minute winter	S43	2.002	S40	87.4	1.257	0.928	2.8340	
30 minute winter	S37	3.000	S38	43.2	1.127	0.332	2.9487	
30 minute winter	S38	3.001	S39	86.2	1.560	0.595	0.9602	
30 minute winter	S39	3.002	S40	117.1	1.663	0.803	2.5853	
30 minute winter	S40	2.003	S17	156.8	2.637	0.944	4.1939	
30 minute winter	S17	1.005_1	S18	307.2	2.185	0.805	4.2865	
30 minute winter	S18	1.006_1	S19	312.1	1.970	0.986	4.1328	
15 minute winter	S35	4.000	S36	41.2	1.363	0.341	3.1498	
15 minute winter	S36	4.001	S19	82.5	1.086	0.290	1.2063	
30 minute winter	S19	1.007_1	S20	359.3	2.356	1.134	7.7135	
30 minute winter	S34	5.000_1	S20	28.4	0.808	0.754	1.7932	
30 minute winter	S20	1.008_1	S21	392.5	2.478	0.637	2.7004	
30 minute winter	S21	1.009_1	S22	412.0	2.601	1.096	1.7386	
30 minute winter	S22	1.010_1	S23	431.5	2.723	1.168	4.4942	
30 minute winter	S23	1.011_1	S24	462.6	2.920	1.222	2.0721	

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Results for 100 year +40% CC Critical Storm Duration. Lowest mass balance: 99.78%

Node Event	US Node	Peak (mins)	Level (m)	Depth (m)	Inflow (l/s)	Node Vol (m³)	Flood (m³)	Status
30 minute winter	S24	23	94.431	1.739	496.1	3.0735	0.0000	SURCHARGED
30 minute winter	S25	22	93.742	2.022	520.0	3.5736	0.0000	FLOOD RISK
30 minute winter	S26	23	92.817	1.997	523.8	5.0819	0.0000	FLOOD RISK
30 minute winter	S27	22	92.753	2.090	559.0	5.3191	0.6137	FLOOD
30 minute winter	S28	22	92.615	2.082	595.9	5.2994	0.0000	SURCHARGED
30 minute winter	S29	22	92.466	1.999	666.2	5.0868	0.0000	SURCHARGED
30 minute winter	S30	22	92.266	1.869	737.0	4.7568	0.0000	SURCHARGED
30 minute winter	S31	22	91.992	1.752	807.8	4.4596	0.0000	SURCHARGED
30 minute winter	S32	22	91.820	1.652	879.0	4.2050	0.0000	SURCHARGED
30 minute winter	S44	23	94.875	0.274	336.4	0.3921	0.0000	OK
30 minute winter	S45	23	93.553	0.823	419.4	0.9310	0.0000	SURCHARGED
30 minute winter	S46	22	92.381	1.151	493.3	1.3021	0.0000	SURCHARGED
30 minute winter	S33	22	91.709	1.559	1382.9	3.9675	0.0000	SURCHARGED
30 minute winter	S47	22	91.431	1.322	1441.8	1.8913	0.0000	SURCHARGED
30 minute winter	S48	22	91.089	1.032	1460.8	1.1673	0.0000	SURCHARGED
30 minute winter	S49	23	90.737	0.746	1460.4	1.0670	0.0000	OK
240 minute winter	S58	240	91.245	0.725	25.2	1.0377	0.0000	OK
240 minute winter	S61	244	91.245	0.769	22.2	1.1010	0.0000	OK
240 minute winter	S59 Flow Control	248	91.244	0.869	46.6	1.2442	0.0000	SURCHARGED
30 minute winter	S60	24	90.673	0.778	1635.1	1.1139	0.0000	OK
30 minute winter	S50	31	90.395	0.633	1793.4	0.9064	0.0000	OK
60 minute winter	S55	47	90.377	0.649	1660.4	0.9284	0.0000	OK
60 minute winter	S51	48	90.360	0.680	1618.9	0.9724	0.0000	OK
60 minute winter	S52	48	90.346	0.773	1568.9	1.1059	0.0000	OK
60 minute winter	S56	48	90.337	0.784	1477.3	1.1222	0.0000	OK

Link Event (Upstream Depth)	US Node	Link	DS Node	Outflow (l/s)	Velocity (m/s)	Flow/Cap	Link Vol (m³)	Discharge Vol (m³)
30 minute winter	S24	1.012_1	S25	486.5	2.443	0.849	15.0527	
30 minute winter	S25	1.013_1	S26	498.7	2.354	1.119	17.2472	
30 minute winter	S26	1.014_1	S27	528.1	1.400	0.435	7.1572	
30 minute winter	S27	1.015_1	S28	562.7	1.279	0.905	22.4035	
30 minute winter	S28	1.016_1	S29	598.8	1.361	1.308	20.8705	
30 minute winter	S29	1.017	S30	668.3	1.519	1.471	22.4677	
30 minute winter	S30	1.018_1	S31	739.0	1.679	1.619	26.0833	
30 minute winter	S31	1.019	S32	809.7	1.519	1.299	24.5705	
30 minute winter	S32	1.020	S33	880.3	1.651	1.596	7.8243	
30 minute winter	S44	6.000	S45	336.3	3.089	0.646	9.4008	
30 minute winter	S45	6.001	S46	415.3	2.942	0.806	9.4095	
30 minute winter	S46	6.002	S33	493.7	3.445	0.818	2.8827	
30 minute winter	S33	1.021	S47	1383.4	2.183	2.017	21.6028	
30 minute winter	S47	1.022	S48	1441.6	2.275	2.067	26.5082	
30 minute winter	S48	1.023	S49	1460.4	2.334	1.719	21.4837	
30 minute winter	S49	1.024	S72	1448.8	0.622	0.214	121.5711	
240 minute winter	S58	7.000	S59 Flow Control	11.5	0.104	0.001	206.7505	
240 minute winter	S61	8.000	S59 Flow Control	16.7	0.128	0.002	142.5447	
240 minute winter	S59 Flow Control	Hydro-Brake®	S60	2.6				
30 minute winter	S60	1.025	S70	1634.9	0.626	0.236	32.7322	
30 minute winter	S50	1.026	S55	1738.0	1.066	0.255	63.2688	
60 minute winter	S55	1.027	S51	1618.9	1.071	0.238	95.2729	
60 minute winter	S51	1.028	S52	1506.0	0.927	0.222	248.4041	
60 minute winter	S52	1.029	S56	1477.3	0.803	0.215	50.7436	
60 minute winter	S56	1.030	S57	1444.1	0.739	0.214	132.9475	

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Results for 100 year +40% CC Critical Storm Duration. Lowest mass balance: 99.78%

Node Event	US Node	Peak (mins)	Level (m)	Depth (m)	Inflow (l/s)	Node Vol (m³)	Flood (m³)	Status
30 minute winter	S57	33	90.329	0.824	1452.4	1.1789	0.0000	OK
30 minute winter	S53	33	90.323	0.883	1418.0	1.2630	0.0000	OK
240 minute winter	S54 HW	252	90.301	1.073	1043.8	3.0406	0.0000	OK
240 minute winter	S62 HW	252	90.301	1.076	542.2	1.5391	0.0000	FLOOD RISK
240 minute winter	S93	252	90.300	0.826	243.5	0.0000	0.0000	FLOOD RISK
15 minute winter	S73	10	91.846	1.373	80.4	1.9648	4.2290	FLOOD
15 minute winter	S74	11	91.747	1.495	182.2	2.1397	0.0000	SURCHARGED
30 minute winter	S71	19	91.325	1.228	232.8	1.7569	0.0000	SURCHARGED
30 minute winter	S72	23	90.729	0.786	1666.1	0.0000	0.0000	OK
15 minute winter	S75	10	92.104	1.651	65.4	2.3622	0.0000	FLOOD RISK
15 minute winter	S76	10	91.893	1.582	175.3	2.2638	1.7987	FLOOD
15 minute winter	S77	10	91.941	1.554	65.4	2.2238	2.3543	FLOOD
30 minute winter	S68	19	91.032	0.853	192.9	1.2204	0.0000	SURCHARGED
30 minute winter	S69	20	90.853	0.820	241.4	1.1739	0.0000	SURCHARGED
30 minute winter	S70	24	90.667	0.785	1800.0	0.0000	0.0000	OK
30 minute winter	S65	25	90.527	0.226	35.0	0.3240	0.0000	OK
30 minute winter	S66	25	90.504	0.396	70.3	0.5672	0.0000	SURCHARGED
30 minute winter	S67	26	90.424	0.593	100.7	0.8485	0.0000	SURCHARGED
30 minute winter	S62	28	90.437	0.617	26.9	0.8825	0.0000	SURCHARGED
30 minute winter	S63	28	90.427	0.647	78.2	0.9253	0.0000	SURCHARGED
30 minute winter	S64	28	90.447	0.577	26.9	0.8255	0.0000	SURCHARGED

Link Event (Upstream Depth)	US Node	Link	DS Node	Outflow (l/s)	Velocity (m/s)	Flow/Cap	Link Vol (m³)	Discharge Vol (m³)
30 minute winter	S57	1.031	S53	1418.0	0.621	0.183	151.5896	
30 minute winter	S53	1.032	S54 HW	1412.3	2.144	1.524	15.4850	
240 minute winter	S54 HW	Flow through pond	S62 HW	542.2	0.107	0.042	5058.0513	
240 minute winter	S62 HW	20.000	Basin A Flow Control	40.5	0.282	0.023	8.6850	
240 minute winter	S93	16.014	S54 HW	240.7	1.561	0.876	4.2645	
15 minute winter	S73	7.000_1	S74	54.6	0.784	0.762	3.7005	
15 minute winter	S74	7.001_1	S71	178.6	1.619	1.586	5.3430	
30 minute winter	S71	7.002	S72	231.3	2.098	2.060	5.3320	
30 minute winter	S72	1.025_1	S60	1632.5	0.637	0.581	126.1596	
15 minute winter	S75	11.000	S76	63.0	0.894	0.989	3.0102	
15 minute winter	S76	11.001	S68	159.0	2.258	2.489	2.7807	
15 minute winter	S77	12.000	S76	49.4	0.701	0.773	1.6028	
30 minute winter	S68	11.002	S69	190.9	1.205	1.029	6.9582	
30 minute winter	S69	11.003	S70	239.6	1.534	1.286	7.1323	
30 minute winter	S70	1.027_1	S50	1793.4	0.858	0.638	260.6319	
30 minute winter	S65	13.000	S66	35.3	0.994	0.491	1.7853	
30 minute winter	S66	13.001	S67	67.0	0.959	0.596	6.9851	
30 minute winter	S67	13.002	S55	99.7	0.852	0.931	14.5351	
30 minute winter	S62	15.000	S63	25.5	0.371	0.533	1.4898	
30 minute winter	S63	14.001	S52	76.7	1.112	1.301	1.9754	
30 minute winter	S64	14.000	S63	26.8	0.406	0.569	3.4477	

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